FUNDAÇÃO GETULIO VARGAS
ESCOLA DE MATEMÁTICA APLICADA – FGV/EMAp
CURSO DE GRADUAÇÃO EM MATEMÁTICA APLICADA

Cálculo Funcional de Itô e Equações Diferenciais Estocásticas Dependentes do Caminho

por
Jeann da Rocha Silva

Rio de Janeiro
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1 Introduction

The Itô's Stochastic Calculus is one of the central tools of modern theory of stochastic process. Your relevance follows from the possibility to analyze quantitatively systems subject to random fluctuations, both in purely probabilistic problems and in applications to physics, biology, engineering, control and mathematical finances. In your classic form, the theory is formulated to functions of type $f : [0, \infty) \times \mathbb{R}^d \to \mathbb{R}$ that are evaluated along of a stochastic process $(X_t)_{t\geq 0}$. In this case, the Itô's formula describes the dynamics of $f(t, X_t)$ in terms of usual derivatives of f and the coefficients of the dynamic of X.

This formulation, while extremely powerful, is adapted to the Markovian case, in which the relevant information at any given moment is encoded by the process's current state. In many problems, however, this assumption is insufficient. There are situations where the observed quantity depends on the process's entire past trajectory rather than just its present value. This occurs, for example, with path-dependent financial derivatives, systems with memory, and control problems where history influences decision-making.

When dependence on the past is essential, the functionals of interest are no longer functions of (t, X_t) but rather functionals of the path up to time t. In other words, instead of considering objects of the form $f(t, X_t)$, one studies expressions of the type $F(X_t)$ where X_t now represents the trajectory of the process up to time t (notice that the information of the time t is embedding in X_t). In this context, standard derivatives with respect to the state are no longer sufficient to formulate an appropriate stochastic chain rule. Thus, the need arises for a differential calculus on path spaces.

Itô Functional Calculus, developed in works by Dupire (see [4]) and subsequently systematized by Cont and Fournié (see [2] and [3]), precisely addresses this need. The central idea involves introducing notions of differentiation adapted to non-anticipative functionals. Two directions of variation are fundamental: the temporal evolution of the functional while the path remains frozen at the current time, and the vertical perturbation of the path's terminal value. These operations lead, respectively, to the horizontal and vertical derivatives of the functional. Under suitable regularity assumptions, these derivatives allow for the formulation of an Itô Functional Formula that generalizes the classical formula.

Beyond its intrinsic interest, this formalism provides the appropriate language for the study of path-dependent stochastic differential equations. Such equations generalize

standard stochastic differential equations by allowing their coefficients to depend on the past trajectory of the process. Consequently, many classical analytical constructions—including the generator, the adjoint operator, and the associated evolution equations—must be reformulated within a functional framework.

The aim of this work is to present the fundamental elements of this theoretical framework. We begin by recalling concepts and results from probability theory (see [5]) and classical stochastic calculus (see [7]) that are necessary to establish notation and context. Next, we introduce path spaces and non-anticipative functionals, define the horizontal and vertical derivatives, state and proof the Itô Functional Formula. We discuss the formulation of path-dependent stochastic differential equations and the interpretation of the associated operators, highlighting the transition from the classical Markovian framework to the functional framework. A preencher...

2 Preliminaries of Probability Theory

2.1 Probability Spaces

The foundation of probability theory is the probability space, which formalizes the notion of a random experiment. All stochastic processes and random variables considered in this work are defined on a common underlying probability space.

A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where

1. Ω is a non-empty set, called the sample space;
2. $\mathcal{F}$ is a σ-algebra on Ω, whose elements are called events;
3. $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a probability measure, i.e., a measure with $\mathbb{P}(\Omega) = 1$.

Given a probability space, a random variable X is a measurable function $X : \Omega \to \mathbb{R}^d$, meaning that for every Borel set $B \subset \mathbb{R}^d$, the preimage $X^{-1}(B)$ belongs to $\mathcal{F}$. The expectation (or expected value) of a random variable X is defined by the Lebesgue integral

$$\mathbb{E}[X] = \int_\Omega X(\omega)\, d\mathbb{P}(\omega)$$

whenever this integral exists.

A central tool in stochastic analysis is the conditional expectation. It allows us to update our probabilistic forecast given partial information represented by a sub-σ-algebra. Let X be an integrable random variable and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra. The conditional expectation of X given $\mathcal{G}$, denoted by $\mathbb{E}[X \mid \mathcal{G}]$, is the unique (up to almost surely equivalence) $\mathcal{G}$-measurable integrable random variable satisfying

$$\mathbb{E}\left[\mathbb{E}[X \mid \mathcal{G}]\mathbb{1}_A\right] = \mathbb{E}\left[X\mathbb{1}_A\right], \qquad \forall A \in \mathcal{G}$$

We will frequently use the following elementary but essential properties of conditional expectation

Proposition 2.1.1. *Let X, Y random variables and $\mathcal{H}, \mathcal{G}, \mathcal{F}$ σ-algebras on Ω. Then,*

1. $\mathbb{E}[\alpha X + \beta Y \mid \mathcal{G}] = \alpha\mathbb{E}[X \mid \mathcal{G}] + \beta\mathbb{E}[Y \mid \mathcal{G}]$ *almost surely.*

2. *If $X \leq Y$ almost surely, then $\mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[Y \mid \mathcal{G}]$ almost surely.*

3. *(Tower property) If $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$, then $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}]$ almost surely.*

4. *If Y is $\mathcal{G}$-measurable and bounded, then $\mathbb{E}[XY \mid \mathcal{G}] = Y\mathbb{E}[X \mid \mathcal{G}]$ almost surely.*

5. *(Jensen's inequality) If $\varphi : \mathbb{R} \to \mathbb{R}$ is convex and X is integrable, then we have $\varphi(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[\varphi(X) \mid \mathcal{G}]$ almost surely.*

In stochastic calculus, the information available up to time t is modeled by a filtration. A filtration is a family $(\mathcal{F}_t)_{t\geq 0}$ of sub-σ-algebras of $\mathcal{F}$ such that $\mathcal{F}_s \subseteq \mathcal{F}_t$ whenever $s \leq t$. The quadruple $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ is called a filtered probability space.

The filtration represents the flow of information over time. A process is said to be adapted to $(\mathcal{F}_t)$ if its value at time t depends only on the information available up to that time (this will be formalized in the next section).

To avoid certain pathologies in stochastic analysis, we will always assume that the filtration satisfies the usual conditions:

1. Completeness: $\mathcal{F}_0$ contains all $\mathbb{P}$-null sets of $\mathcal{F}$ (and hence so does every $\mathcal{F}_t$);

2. Right-continuity: $\mathcal{F}_t = \mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$ for all $t \geq 0$.

These conditions guarantee that stopping times have good properties and that every martingale has a càdlàg modification, which is essential for the rigorous construction of stochastic integrals. Speaking of which, a random variable $\tau : \Omega \to [0, \infty]$ is a stopping time with respect to $(\mathcal{F}_t)_{t\geq 0}$ if

$$\{\tau \leq t\} \in \mathcal{F}_t, \qquad \forall t \geq 0$$

Intuitively, a stopping time is a random time at which we can decide whether it has occurred or not based solely on the information available up to that time.

A stochastic process is a collection of random variables indexed by time. In this work, we focus on continuous-time processes, which are the main objects of study in stochastic calculus.

A stochastic process is a family $(X_t)_{t\geq 0}$ of random variables $X_t : \Omega \to \mathbb{R}^d$, all defined on the same probability space. The map $t \mapsto X_t(\omega)$, for fixed $\omega \in \Omega$, is called a sample path or trajectory of the process.

We now formalize the notion of non-anticipating processes, which are the admissible integrands in stochastic integration. A stochastic process $(X_t)_{t\geq 0}$ is adapted to the filtration $(\mathcal{F}_t)_{t\geq 0}$ if X_t is $\mathcal{F}_t$-measurable for every $t \geq 0$.

A stronger measurability condition, which is crucial for the construction of stochastic integrals, is progressive measurability. A stochastic process $(X_t)_{t\geq 0}$ is progressively measurable with respect to $(\mathcal{F}_t)_{t\geq 0}$ if, for every $t \geq 0$, the map

$$(s, \omega) \in [0, t] \times \Omega \longmapsto X_s(\omega) \in \mathbb{R}^d$$

is measurable with respect to the product σ-algebra $\mathcal{B}([0,t]) \otimes \mathcal{F}_t$, where $\mathcal{B}([0,t])$ denotes the Borel σ-algebra on $[0,t]$.

Progressive measurability is stronger than adaptedness. However, under the usual conditions, every adapted process with right-continuous (or left-continuous) paths is progressively measurable. This property is indispensable when defining stochastic integrals, as it ensures that the integrand does not anticipate future increments of the integrator.

For the general theory of stochastic integration that we will see we need a notion that is even stronger in terms of "non-anticipation" than progressive measurability: the concept of predictable processes. The predictable σ-algebra captures the idea of events that are "known just before" they happen.

The predictable σ-algebra $\mathcal{P}$ on $\Omega \times [0,\infty)$ is the σ-algebra generated by the following families of sets:

1. $\{0\} \times A$, for $A \in \mathcal{F}_0$;
2. $(s,t] \times A$, for $0 \leq s < t$ and $A \in \mathcal{F}_s$.

A process $(X_t)_{t\geq 0}$ is called predictable if the map $(t,\omega) \mapsto X_t(\omega)$ is $\mathcal{P}$-measurable.

Equivalently, the predictable σ-algebra is generated by all left-continuous adapted processes. This means that a predictable process is one whose value at time t can be determined from the information available strictly before time t.

The relationships between the different measurability classes are:

$$\text{Predictable} \subset \text{Progressively Measurable} \subset \text{Adapted}.$$

For continuous processes (or processes with left-continuous paths), adaptedness implies predictability (since left-continuous adapted processes are predictable), so these distinctions often disappear.

Since we will frequently deal with processes defined up to almost sure equivalence, it is useful to distinguish between different notions of equality for stochastic processes.

Let $(X_t)_{t\geq 0}$ and $(Y_t)_{t\geq 0}$ be two stochastic processes.

1. Y is a modification of X if, for every $t \geq 0$, $\mathbb{P}(X_t = Y_t) = 1$.
2. X and Y are indistinguishable if $\mathbb{P}(X_t = Y_t \text{ for all } t \geq 0) = 1$.

Remark 2.1.2. *For processes with right-continuous or left-continuous paths, modification implies indistinguishability. Indeed, if two modifications have continuous paths, they are indistinguishable because they agree on a countable dense subset of* $[0,\infty)$. *This fact will be used repeatedly in the construction of the continuous version of the stochastic integral.*

2.2 Martingales

Martingales are stochastic processes that model fair games: given the past information, the expected future value equals the present value. They provide the fundamental probabilistic structure underlying stochastic integration and the Itô's formula.

Throughout this chapter, we consider a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions (i.e., the filtration is right-continuous and complete). All processes are assumed to be adapted to this filtration.

An adapted process $(M_t)_{t\geq 0}$ with values in $\mathbb{R}$ is a martingale if

1. $\mathbb{E}[|M_t|] < \infty$ for all $t \geq 0$;
2. for all $0 \leq s \leq t$, $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ almost surely.

If $\mathbb{E}[M_t \mid \mathcal{F}_s] \geq M_s$ (respectively $\leq M_s$), it is called a submartingale (respectively supermartingale).

A key property of martingales is that they have constant expectation: $\mathbb{E}[M_t] = \mathbb{E}[M_0]$ for all $t \geq 0$. This contrasts with local martingales, which may have non-constant expectation in general.

An adapted process $(M_t)_{t\geq 0}$ in $\mathbb{R}$ is a local martingale if there exists a sequence of stopping times $(\tau_n)_{n\in\mathbb{N}}$ such that $\tau_n \uparrow \infty$ almost surely and, for each n, the stopped process $(M_{t\wedge\tau_n})_{t\geq 0}$ is a martingale.

A local martingale that is also bounded below (or bounded above) is a supermartingale (or submartingale). Every martingale is a local martingale, but the converse is not true without additional integrability conditions.

For vector-valued processes $M_t = (M_t^{(1)}, \ldots, M_t^{(d)})$, we say that $(M_t)_{t\geq 0}$ is a (local) martingale if each component $M_t^{(i)}$ is a (local) martingale. This componentwise definition is standard and suffices for all applications in this work.

We now recall an essential probabilistic estimate used extensively in the construction and analysis of stochastic integrals.

Proposition 2.2.1 (Doob's Maximal Inequality)**.** *Let $(M_t)_{t\in[0,T]}$ be a continuous martingale (or a non-negative submartingale). For any $p \geq 1$ and $\varepsilon > 0$,*

$$\mathbb{P}\left(\sup_{0\leq t\leq T} \|M_t\|_{\mathbb{R}^m} \geq \varepsilon\right) \leq \frac{1}{\varepsilon^p}\mathbb{E}\left[\|M_T\|_{\mathbb{R}^m}^p\right]$$

Another important result is the Optional Stopping Theorem, which allows one to replace fixed times by bounded stopping times in the martingale identity.

We end this section with some considerations about the variation of a process. More precisely the total variation and quadratic variation. In special, the quadratic variation

measures the cumulative variability of a stochastic process and is the central object that distinguishes stochastic calculus from classical calculus.

For a real-valued continuous process $(X_t)_{t\geq 0}$, its total variation and quadratic variation are, respectively, the process $(|X|_t)_{t\geq 0}$ and $(\langle X\rangle_t)_{t\geq 0}$ defined (the second if the limit exists in probability) by

$$|X|_t = \sup_{\pi} \sum_{k=1}^{n} \left|X_{t_k} - X_{t_{k-1}}\right| \qquad \text{and} \qquad \langle X\rangle_t = \lim_{\|\pi\|\to 0} \sum_{k=1}^{n} \left(X_{t_k} - X_{t_{k-1}}\right)^2$$

where $\pi = (t_k)_{k=1}^n$ is any partition of $[0,t]$, and $\|\pi\| = \max_{1\leq k\leq n}(t_k - t_{k-1})$. If $(\pi_n)_n$ is a sequence of partitions of $[0,t]$ such that $\|\pi_n\| \to 0$ we say that π_n is a sequence of partitions converging to identity.

When dealing with vector-valued local martingales, the cross-variation (or covariation) between two components becomes essential. For two real-valued continuous semimartingales $(X_t)_{t\in[0,T]}$ and $(Y_t)_{t\in[0,T]}$, their covariation (or cross-variation) is defined by polarization:

$$\langle X, Y\rangle_t = \frac{1}{4}\left(\langle X+Y\rangle_t - \langle X-Y\rangle_t\right)$$

Equivalently, it is the limit (in probability) of the Riemann sums

$$\langle X, Y\rangle_t = \lim_{\|\pi\|\to 0} \sum_{k=1}^{n} \left(X_{t_k} - X_{t_{k-1}}\right)\left(Y_{t_k} - Y_{t_{k-1}}\right)$$

Let $(M_t)_{t\geq 0} = (M_t^{(1)}, \ldots, M_t^{(d)})_{t\geq 0}$ be a continuous local martingale in $\mathbb{R}^d$. Its quadratic variation is the matrix-valued process $\langle M\rangle_t = \left(\langle M^{(i)}, M^{(j)}\rangle_t\right)_{1\leq i,j\leq d}$, where each entry is the covariation of the corresponding scalar components.

Proposition 2.2.2. *Let $(M_t)_{t\geq 0}$ be a continuous martingale in $\mathbb{R}^d$. The matrix $\langle M\rangle_t$ has the following properties:*

1. *It is symmetric: $\langle M^{(i)}, M^{(j)}\rangle_t = \langle M^{(j)}, M^{(i)}\rangle_t$.*
2. *It is non-decreasing: $\langle M^{(i)}\rangle_t \geq \langle M^{(i)}\rangle_s$ if $t \geq s$. In particular it is positive: $\langle M^{(i)}\rangle_t \geq 0$.*
3. *It is positive semidefinite almost surely for each $t \geq 0$. Indeed, for any vector $v \in \mathbb{R}^d$, we have $v^\top \langle M\rangle_t v = \langle v \cdot M\rangle_t \geq 0$, since the quadratic variation of the scalar local martingale $v \cdot M$ is non-decreasing.*
4. *The process $\langle M\rangle_t$ is continuous and of finite quadratic variation in each entry. If $(M_t)_{t\geq 0}$ has finite total variation, i.e., $|M|_t < \infty$ for all $t \geq 0$, then its quadratic variation is zero.*
5. *The process $M_t M_t^\top - \langle M\rangle_t$ is a local martingale in the space of symmetric matrices.*

3 Preliminaries of Stochastic Calculus

3.1 Brownian Motion

Brownian motion is one of the fundamental objects in probability theory and stochastic analysis. Beyond its mathematical importance, it has become an essential model in numerous applications, including physics, finance, biology, engineering, and other scientific disciplines.

Historically, the first major applications were proposed independently by Louis Bachelier, in the context of financial markets, and Albert Einstein, in the study of the random motion of microscopic particles suspended in a fluid. These pioneering works demonstrated the relevance of random processes for describing complex natural and economic phenomena.

The rigorous mathematical foundations of Brownian motion were established later by Norbert Wiener, who proved its existence as a stochastic process. For this reason, Brownian motion is also commonly referred to as the Wiener process.

A standard $\mathbb{R}^d$-valued Brownian motion is a stochastic process $(W_t)_{t\geq 0}$ with values in $\mathbb{R}^d$ satisfying:

1. $W_0 = 0$;
2. For any $0 \leq s < t$, the increment $W_t - W_s$ is independent of $\sigma(W_u | 0 \leq u \leq s)$, and has normal distribution $\mathcal{N}(0, (t-s)I_d)$;
3. The paths $t \mapsto W_t$ are continuous almost surely.

We can see that a standard $\mathbb{R}^d$-valued Brownian motion $(W_t)_{t\geq 0}$ is a vector-valued stochastic process $(W_t^{(1)}, ..., W_t^{(d)})_{t\geq 0}$ of independent standard real-valued Brownian motions $(W_t^{(i)})_{t\geq 0}$. Actually, it is immediate from the fact that zero covariance implies independence for jointly Gaussian random variables. Even clearer is the converse: a vector-valued stochastic process of d independent standard real-valued Brownian motion is a standard $\mathbb{R}^d$-valued Brownian motion.

Proposition 3.1.1. *Let $(W_t)_{t\geq 0}$ be a standard $\mathbb{R}^d$-valued Brownian motion. Then*

$$\mathbb{E}[W_t] = 0 \qquad \textit{and} \qquad \textit{Cov}(W_s, W_t) = (s \wedge t) I_d.$$

Proposition 3.1.2. *If $(X_t)_{t\geq 0}$ is a $\mathbb{R}^d$-valued Gaussian process with $X_0 = 0$, $\mathbb{E}[X_t] = 0$, $Cov(X_s, X_t) = (s \wedge t)I_d$ and it has continuous paths almost surely, then it is a standard $\mathbb{R}^d$-valued Brownian motion.*

Proposition 3.1.3. *A standard $\mathbb{R}^d$-valued Brownian motion $(W_t)_{t\geq 0}$ is a martingale with respect to filtration $(\mathcal{F}_t)_{t\geq 0}$ given by $\mathcal{F}_t = \sigma(W_s|0 \leq s \leq t)$.*

Proposition 3.1.4. *If $(W_t)_{t\geq 0}$ is a standard $\mathbb{R}^d$-valued Brownian motion, its quadratic variation is $\langle W\rangle_t = tI_d$ almost surely. Consequently, it has almost surely a infinite total variation, i.e., $|W|_t = \infty$ almost surely.*

Theorem 3.1.5 (Existence of Brownian Motion)**.** *There exists a standard $\mathbb{R}^d$-valued Brownian motion.*

Proof Sketch. One possible construction of a standard real-valued Brownian motion is based on the Haar wavelet basis $(H_n)_n$ in $L^2[0,1]$ where H_n are called wavelets and they are defined via the "mother wavelet" $H : \mathbb{R} \to \mathbb{R}$ given by

$$H(t) = \begin{cases} 1 & \text{if } t \in [0, 1/2] \\ -1 & \text{if } t \in [1/2, 1] \\ 0 & \text{otherwise} \end{cases}$$

as $H_n(t) = 2^{j/2}H(2^j t - k)$ where $n = 2^j + k$, $j \in \mathbb{N}$, $k = 0, ..., 2^j - 1$. The sequence $(H_n)_n$ is a complete orthonormal basis of $L^2[0,1]$.

The idea is to begin with a sequence of independent standard Gaussian random variables $(Z_n)_n$ and use them as the coefficients of the random series

$$X_t = \sum_{n=1}^{\infty} \lambda_n Z_n \Delta_n(t) = \lim_{n\to\infty} \sum_{k=1}^{n} \lambda_k Z_k \Delta_k(t), \quad t \in [0,1] \tag{3.1}$$

where $\Delta_n \in L^2[0,1]$ are called Schauder functions and they are defined via $\Delta : \mathbb{R} \to \mathbb{R}$ given by

$$\Delta(t) = \begin{cases} 2t & \text{if } t \in [0, 1/2] \\ 2(1-t) & \text{if } t \in [1/2, 1] \\ 0 & \text{otherwise} \end{cases}$$

as $\Delta_n(t) = \Delta(2^j t - k)$ where $n = 2^j + k, j \in \mathbb{N}, k = 0, ..., 2^j - 1$. The coefficients $\lambda_n \in \mathbb{R}$ are chosen to satisfy $\lambda_n \Delta_n(t) = \int_0^t H_n(s)\, ds$.

The Brownian sample paths are obtained as the almost surely uniform limit of (3.1). The coefficients Z_n determine the random fluctuations at each dyadic scale, while the functions $\lambda_n \Delta_n$ provide a hierarchical refinement of the paths. Uniform convergence follows from suitable estimates on the Gaussian coefficients together with the rapidly decreasing

size of those functions. The resulting process has continuous sample paths almost surely since the functions $\lambda_n Z_n \Delta_n$ are continuous.

We can see that $\mathrm{Cov}(W_s, W_t) = s \wedge t$ by a direct computation. Furthermore, a computation of the characteristic function of the finite-dimensional distributions shows that this is a Gaussian process. Then, Proposition 3.1.2 implies that the process is a real-valued standard Brownian motion on $[0, 1]$. The construction on $[0, 1]$ can be extended to $[0, \infty)$ by concatenating independent Brownian motions on successive intervals of length one.

A standard $\mathbb{R}^d$-valued Brownian motion is then obtained by taking d independent copies of the one-dimensional construction. □

Two other fascinating facts about Brownian motion are presented below.

Proposition 3.1.6. *The paths of Brownian motion are almost surely α-Hölder continuous with $\alpha < \frac{1}{2}$.*

Proposition 3.1.7. *The paths of Brownian motion are almost surely not α-Hölder continuous with $\alpha \geq \frac{1}{2}$. In particular, they are almost surely not differentiable.*

There are many other interesting facts about Brownian motion. However, if we devote more time to this topic, we will not be able to advance the focus of this work.

3.2 Stochastic Integrals

The stochastic integral generalizes the Riemann–Stieltjes integral to allow integration with respect to Brownian motion. Since Brownian paths are of unbounded variation (cf. Proposition 3.1.4), the classical Riemann–Stieltjes integral does not apply. From this point on, we will always suppose that $(W_t)_{t\geq 0}$ is a standard $\mathbb{R}^d$-valued Brownian motion with respect to some filtration $(\mathcal{F}_t)_{t\geq 0}$.

We begin by defining the stochastic integral for simple processes. We say that a process $(X_t)_{t\in[0,T]}$ taking values in $\mathbb{R}^{m\times d}$ is simple if there exist a partition $(t_k)_{k=1}^n$ of $[0, T]$ and bounded $\mathcal{F}_{t_{k-1}}$-measurable random matrices $A_{k-1} \in \mathbb{R}^{m\times d}$ such that

$$X_t = A_0 \mathbb{1}_{\{0\}}(t) + \sum_{k=1}^{n} A_{k-1} \mathbf{1}_{(t_{k-1}, t_k]}(t), \qquad t \in [0, T].$$

Moreover, for this process, the stochastic integral is defined by

$$I((X_t)_{t\in[0,T]}) = \int_0^T X_t \, dW_t = \sum_{k=1}^{n} A_{k-1}(W_{t_k} - W_{t_{k-1}}),$$

where the products are matrix-vector multiplications, so the result is an element of $\mathbb{R}^m$.

Next, we would like to extend this to a larger class of progressively measurable integrands $(X_t)_{t\in[0,T]}$ in $\mathbb{R}^{m\times d}$ such that

$$\mathbb{E}\left[\int_0^T \|X_t\|^2\, dt\right] < \infty,$$

where $\|X_t\|^2 = \sum_{i=1}^{m}\sum_{j=1}^{d}(X_t^{(i,j)})^2$ denotes the squared Frobenius norm. This extension is possible by the following fundamental isometry.

Proposition 3.2.1 (Itô's Isometry)**.** *For any simple process $(X_t)_{t\in[0,T]}$ in $\mathbb{R}^{m\times d}$ we have*

$$\mathbb{E}\left[\left\|\int_0^T X_t\, dW_t\right\|_{\mathbb{R}^m}^2\right] = \mathbb{E}\left[\int_0^T \|X_t\|^2\, dt\right].$$

Remark 3.2.2. *The space*

$$\mathcal{H}^2[0,T] = \left\{(X_t)_{t\in[0,T]} \text{ progressively measurable in } \mathbb{R}^{m\times d} \;\middle|\; \mathbb{E}\left[\int_0^T \|X_t\|^2\, dt\right] < \infty\right\}$$

with the inner product $\langle (X_t)_{t\in[0,T]}, (Y_t)_{t\in[0,T]}\rangle_{\mathcal{H}^2} = \mathbb{E}\left[\int_0^T \operatorname{tr}\left(X_t Y_t^\top\right) dt\right]$ is a Hilbert space generated by the simple processes.

Proposition 3.2.1 and Remark 3.2.2 give us that the stochastic integral can be extended to the space $\mathcal{H}^2[0,T]$. More precisely, let $(X_t)_{t\in[0,T]} \in \mathcal{H}^2[0,T]$. Then, there is a sequence $(X_t^{(n)})_{t\in[0,T]}$ of simple processes such that $\|(X_t^{(n)})_{t\in[0,T]} - (X_t)_{t\in[0,T]}\|_{\mathcal{H}^2} \to 0$ and we define the stochastic integral of $(X_t)_{t\in[0,T]}$ by $I((X_t)_{t\in[0,T]}) = \lim_{n\to\infty} I((X_t^{(n)})_{t\in[0,T]})$. This limit is well defined.

Proposition 3.2.3. *Let $(X_t)_{t\geq 0}, (Y_t)_{t\geq 0} \in \mathcal{H}^2[0,T]$. Then*

1. $\int_0^T (\alpha X_t + \beta Y_t)\, dW_t = \alpha \int_0^T X_t\, dW_t + \beta \int_0^T Y_t\, dW_t$;
2. *(Itô's Isometry)* $\mathbb{E}\left[\left\|\int_0^T X_t\, dW_t\right\|_{\mathbb{R}^m}^2\right] = \mathbb{E}\left[\int_0^T \|X_t\|^2\, dt\right]$.

The stochastic integral defined above is an element of $L^2(\Omega;\mathbb{R}^m)$ (i.e., I is an operator from $\mathcal{H}^2[0,T]$ to $L^2(\Omega;\mathbb{R}^m)$). In order to define an operator that takes a stochastic process to a stochastic process, we could consider the process $Y_s = I((X_t \mathbb{1}_{[0,s]})_{t\in[0,T]})$, which we denote by $\int_0^s X_t\, dW_t$. However, this is not enough because for each $s \in [0,T]$ we can define $I((X_t\mathbb{1}_{[0,s]})_{t\in[0,T]})$ arbitrarily on a null set $A_s \subset \Omega$. Then, this process is not well defined on $\bigcup_{s\in[0,T]} A_s$, which is not necessarily a null set because $[0,T]$ is uncountable. This issue can be overcome by a versioning argument. The following theorem states this result.

Proposition 3.2.4. *Let $(X_t)_{t\in[0,T]} \in \mathcal{H}^2[0,T]$. Then there exists a continuous martingale $(Y_t)_{t\in[0,T]}$ with values in $\mathbb{R}^m$ such that $\mathbb{P}\left(Y_t = \int_0^t X_s\, dW_s,\ \forall t \in [0,T]\right) = 1$.*

Proof Sketch. The idea is to approximate $(X_t)_{t\in[0,T]}$ by a sequence of simple processes $(X_t^{(n)})_{t\in[0,T]}$ such that

$$\mathbb{E}\left[\int_0^T \|X_t^{(n)} - X_t\|^2\, dt\right] = \|(X_t^{(n)})_{t\in[0,T]} - (X_t)_{t\in[0,T]}\|_{\mathcal{H}^2} \to 0$$

For each simple process $(X_t^{(n)})_{t\in[0,T]}$, it can be seen by the initial definition that the process $Y_s^{(n)} = \int_0^s X_t^{(n)}\, dW_t$ is a continuous martingale.

The Itô's isometry together with Doob's maximal inequality gives

$$\begin{aligned}\mathbb{P}\left(\sup_{0\le t\le T} \|Y_t^{(n)} - Y_t^{(m)}\|_{\mathbb{R}^m} \ge \varepsilon\right) &\le \frac{1}{\varepsilon^2}\mathbb{E}\left[\|Y_T^{(n)} - Y_T^{(m)}\|_{\mathbb{R}^m}^2\right] \\ &= \frac{1}{\varepsilon^2}\mathbb{E}\left[\int_0^T \|X_t^{(n)} - X_t^{(m)}\|^2\, dt\right]\end{aligned}$$

By choosing a suitable subsequence $(Y_t^{(n_k)})_{t\in[0,T]}$ and applying the Borel-Cantelli lemma, we obtain almost surely that this is a Cauchy sequence in $C([0,T];\mathbb{R}^m)$ and then a uniform convergence on $[0,T]$ to a continuous process $(Y_t)_{t\in[0,T]}$. Since each $(Y_t^{(n)})_{t\in[0,T]}$ is a martingale, passing to the limit in the martingale identity shows that $(Y_t)_{t\in[0,T]}$ is also a martingale.

Finally, the Itô's isometry implies that the limit does not depend on the approximating sequence and satisfies $Y_s = \int_0^s X_t\, dW_t$ almost surely for each $s \in [0,T]$. □

The definition of the stochastic integral on $\mathcal{H}^2[0,T]$ requires the strong condition that $\mathbb{E}\left[\int_0^T \|X_t\|^2\, dt\right] < \infty$. However, many integrands of interest — such as continuous functions of Brownian motion that may grow rapidly, for instance, $f(W_t) = \exp(W_t^4)$ — fail to satisfy this condition, even though $\int_0^t \|X_s\|^2\, ds$ is finite almost surely for each $t \in [0,T]$. To cover these cases, we introduce the class

$$\mathcal{L}^2_{\text{loc}}[0,T] = \left\{ (X_t)_{t\in[0,T]} \text{ progressively measurable in } \mathbb{R}^{m\times d} \,\middle|\, \int_0^T \|X_t\|^2\, dt < \infty\ \mathbb{P}\text{- a.s.} \right\}$$

Equivalently, there exists a sequence of stopping times $(\tau_n)_n$ with $0 \le \tau_n \le T$ and $\tau_n \uparrow T$ almost surely, such that the stopped process $X_t^{\tau_n} = X_t \mathbb{1}_{[0,\tau_n]}$ belongs to $\mathcal{H}^2[0,T]$ for each $n \in \mathbb{N}$. A canonical example of such a sequence is

$$\tau_n = \inf\left\{ t \in [0,T] \,\middle|\, \int_0^t \|X_s\|^2\, ds \ge n \right\} \wedge T$$

which is a stopping time (provided the usual conditions on the filtration and the process is

progressively measurable) and, by construction, satisfies

$$\mathbb{E}\left[\int_0^T \|X_t^{\tau_n}\|^2 \, dt\right] \leq n < \infty$$

thus guaranteeing that $(X_t^{\tau_n})_{t\in[0,T]} \in \mathcal{H}^2[0,T]$.

For $(X_t)_{t\in[0,T]} \in \mathcal{L}^2_{\text{loc}}[0,T]$, we define the stochastic integral by localization. For each $n \in \mathbb{N}$, the previous proposition guarantees the existence of the continuous martingale $Y_s^{(n)} = \int_0^s X_t^{\tau_n} \, dW_t$. By the uniqueness of the integral, we have the consistency condition $Y_t^{(n+1)} = Y_t^{(n)}$ for all $t \leq \tau_n$. Therefore, the paths of the processes $(Y_t^{(n)})_{t\in[0,T]}$ glue together uniquely, defining a continuous process $(Y_t)_{t\in[0,T]}$ such that $Y_t = Y_t^{(n)}$ whenever $t \leq \tau_n$. We then write $Y_s = \int_0^s X_t \, dW_t$.

The process $(Y_t)_{t\in[0,T]}$ thus obtained is not necessarily a square-integrable martingale, but rather a continuous local martingale. Indeed, for each $n \in \mathbb{N}$, the stopped process $Y_{t\wedge\tau_n} = Y_{t\wedge\tau_n}^{(n)}$ is a martingale (continuous and square-integrable). This notion of a local martingale is precisely the structure needed to extend the integral to more general integrands and, as we shall see next, to define integration with respect to semimartingales.

For more general integrators (instead of dW_t), we need the concept of a semimartingale. An adapted process in $\mathbb{R}^d$ $(X_t)_{t\geq 0}$ is a semimartingale if it can be decomposed as

$$X_t = X_0 + M_t + A_t \tag{3.2}$$

where $(M_t)_{t\geq 0}$ is a local martingale in $\mathbb{R}^d$ and $(A_t)_{t\geq 0}$ is a predictable finite-variation process in $\mathbb{R}^d$. It is immediate that Brownian motion is a semimartingale.

More generally, if $(X_t)_{t\in[0,T]}$ is a continuous semimartingale in $\mathbb{R}^d$ with decomposition (3.2) and $(Y_t)_{t\in[0,T]}$ is a progressively measurable process in $\mathbb{R}^{m\times d}$ satisfying the local integrability conditions

$$\int_0^t \|Y_s\| \, d|A|_s < \infty \quad \text{a.s.}$$

and

$$\int_0^t Y_s \, d\langle M\rangle_s \, Y_s^\top < \infty \quad \text{componentwise a.s.}$$

for all $t \in [0,T]$, then the stochastic integral $\int_0^s Y_t \, dX_t$ is well-defined as

$$\int_0^s Y_t \, dX_t = \int_0^s Y_t \, dM_t + \int_0^s Y_t \, dA_t$$

where the first term is the stochastic integral with respect to the local martingale M (constructed via an isometry and localization such as in the Brownian case), and the second is

the pathwise Lebesgue-Stieltjes integral, defined componentwise:

$$\left(\int_0^s Y_t\, dA_t\right)_i = \sum_{j=1}^{d} \int_0^s Y_t^{(i,j)}\, dA_t^{(j)}, \qquad i = 1, \ldots, m$$

Finally, for processes with jumps, i.e., semimartingales non necessarely continuous, we require that $(Y_t)_{t\in[0,T]}$ must be predictable to ensure that the integral does not "see" the jump of X at the same instant, which would violate the principle of non-anticipation.

Remark 3.2.5. *When we are integrating with respect to Brownian motion, the stochastic integral is also called the Itô's integral.*

Proposition 3.2.6. *Let $(X_t)_{t\geq 0}$ be a semimartingale in $\mathbb{R}^d$ with decomposition (3.2) and let $(Y_t)_{t\geq 0}, (Z_t)_{t\geq 0}$ predictable processes in $\mathbb{R}^{m\times d}$ satisfying the local integrability conditions described above. Then,*

1. $\int_0^T (\alpha Y_t + \beta Z_t)\, dX_t = \alpha \int_0^T Y_t\, dX_t + \beta \int_0^T Z_t\, dX_t$;
2. *(Itô's isometry)* $\mathbb{E}\left[\left\|\int_0^T Y_t\, dM_t\right\|_{\mathbb{R}^m}^2\right] = \mathbb{E}\left[\int_0^T \text{tr}\left(Y_t\, d\langle X\rangle_t\, Y_t^\top\right)\right]$

3.3 Itô's Formula

Itô's formula is the chain rule of stochastic calculus. It is a cornerstone for deriving functional equations and for solving stochastic differential equations.

Theorem 3.3.1 (Itô's Formula)**.** *Let $(X_t)_{t\in[0,T]}$ be a continuous semimartingale in $\mathbb{R}^d$ and $f \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d; \mathbb{R}^m)$. Then, for each $t \in [0,T]$,*

$$\begin{aligned} f(t, X_t) = f(0, X_0) &+ \int_0^t \partial_s f(s, X_s)\, ds \\ &+ \sum_{i=1}^{d} \int_0^t \partial_{x_i} f(s, X_s)\, dX_s^{(i)} \\ &+ \frac{1}{2} \sum_{i,j=1}^{d} \int_0^t \partial_{x_i x_j} f(s, X_s)\, d\langle X^{(i)}, X^{(j)}\rangle_s \end{aligned} \tag{3.3}$$

where $\partial_{x_i} f \in \mathbb{R}^m$ and $\partial_{x_i x_j} f \in \mathbb{R}^m$.

Remark 3.3.2. *Theorem 3.3.1 can be rewritten in the concise matrix form*

$$\begin{aligned} f(t, X_t) = f(0, X_0) &+ \int_0^t \partial_s f(s, X_s)\, ds + \int_0^t \partial_x f(s, X_s)\, dX_s \\ &+ \frac{1}{2} \int_0^t \text{tr}\left(\partial_{xx} f(s, X_s)\, d\langle X\rangle_s\right) \end{aligned} \tag{3.4}$$

where $\partial_x f \in \mathbb{R}^{m\times d}$ is the Jacobian matrix, $\partial_{xx} f$ is the Hessian tensor (in $\mathbb{R}^{m\times d\times d}$), and the trace is taken over the spatial indices $1, \dots, d$. Componentwise,

$$\left(\operatorname{tr}\left(\partial_{xx} f(s, X_s)\, d\langle X\rangle_s\right)\right)_\ell = \sum_{i,j=1}^{d} \partial_{x_i x_j} f_\ell(s, X_s)\, d\langle X^{(i)}, X^{(j)}\rangle_s \qquad \ell = 1, ..., m.$$

A $\mathbb{R}^d$-valued Itô's process is a stochastic process of the form

$$X_t = X_0 + \int_0^t \mu_s\, ds + \int_0^t \sigma_s\, dW_s$$

where $(W_t)_{t\geq 0}$ is a standard Brownian motion in $\mathbb{R}^k$, $(\mu_t)_{t\geq 0}$ is a progressively measurable process in $\mathbb{R}^d$ with $\int_0^t \|\mu_s\|\, ds < \infty$ a.s., and $(\sigma_t)_{t\geq 0}$ is a progressively measurable process in $\mathbb{R}^{d\times k}$ with $\int_0^t \|\sigma_s\|^2\, ds < \infty$ a.s. for every $t \geq 0$. In differential notation,

$$dX_t = \mu_t\, dt + \sigma_t\, dW_t$$

If $(X_t)_{t\geq 0}$ is an Itô's process and $f \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d; \mathbb{R}^m)$ (or $C^2(\mathbb{R}^d; \mathbb{R}^m)$ in the time-homogeneous case), then Itô's formula takes the compact form

$$df(t, X_t) = \partial_t f(t, X_t)\, dt + \partial_x f(t, X_t)\, dX_t + \frac{1}{2} \operatorname{tr}\left(\partial_{xx} f(t, X_t)\, d\langle X\rangle_t\right)$$

Expanding dX_t and $d\langle X\rangle_t = \sigma_t \sigma_t^\top dt$, we obtain

$$\begin{aligned} df(t, X_t) = \Big(&\partial_t f(t, X_t) + \partial_x f(t, X_t)\, \mu_t \\ &+ \frac{1}{2} \operatorname{tr}\left(\partial_{xx} f(t, X_t)\, \sigma_t \sigma_t^\top\right)\Big) dt + \partial_x f(t, X_t)\, \sigma_t\, dW_t \end{aligned}$$

Here, the products are understood in the usual matrix–vector sense:

- $\partial_x f \in \mathbb{R}^{m\times d}$, $\mu_t \in \mathbb{R}^d \Rightarrow \partial_x f\ \mu_t \in \mathbb{R}^k$;
- $\partial_x f \in \mathbb{R}^{m\times d}$, $\sigma_t \in \mathbb{R}^{d\times k} \Rightarrow \partial_x f\ \sigma_t \in \mathbb{R}^{m\times k}$, multiplied by $dW_t \in \mathbb{R}^k$ yields $\mathbb{R}^m$;
- The trace term is computed componentwise as above.

3.4 Stochastic Differential Equations

A stochastic differential equation (SDE) has the form

$$dX_t = b(t, X_t)\, dt + \sigma(t, X_t)\, dW_t, \qquad X_0 = \xi$$

where $b : [0, \infty) \times \mathbb{R}^d \to \mathbb{R}^d$ and $\sigma : [0, \infty) \times \mathbb{R}^d \to \mathbb{R}^{d\times k}$ are jointly measurable functions. A strong solution is an adapted process $(X_t)_{t\geq 0}$ with almost surely continuous paths such

that

$$\int_0^t \|b(s, X_s)\|\, ds < \infty \quad \text{a.s.}, \qquad \int_0^t \|\sigma(s, X_s)\|^2\, ds < \infty \quad \text{a.s.}$$

for every $t \geq 0$, and satisfying, for all $t \geq 0$,

$$X_t = \xi + \int_0^t b(s, X_s)\, ds + \int_0^t \sigma(s, X_s)\, dW_s$$

The progressive measurability of $t \mapsto b(t, X_t)$ and $t \mapsto \sigma(t, X_t)$ follows from the fact that $(X_t)_{t\geq 0}$ is adapted and has continuous paths, provided b and σ are Borel measurable.

Theorem 3.4.1 (Existence and Uniqueness)**.** *Assume that the initial condition ξ is $\mathcal{F}_0$-measurable and that there exists a constant $K > 0$ such that for all $x, y \in \mathbb{R}^d$ and $t \geq 0$,*

$$\|b(t, x) - b(t, y)\| + \|\sigma(t, x) - \sigma(t, y)\| \leq K\|x - y\|$$

and

$$\|b(t, x)\| + \|\sigma(t, x)\| \leq K(1 + \|x\|)$$

Then the SDE admits a unique strong solution. In particular, the integrability conditions required for the stochastic integral are automatically satisfied.

Example 3.4.2 (Geometric Brownian Motion)**.** *The SDE*

$$dX_t = \mu X_t\, dt + \sigma X_t\, dW_t, \qquad X_0 = x_0 > 0$$

has the explicit solution

$$X_t = x_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W_t\right)$$

which is obtained by applying Itô's formula to $f(x) = \log x$.

Theorem 3.4.3 (Feynman-Kac's Formula)**.** *Consider the terminal value problem*

$$\begin{cases} \partial_t u(t, x) + \mathcal{L}u(t, x) - r(t, x)u(t, x) = 0, & (t, x) \in [0, T) \times \mathbb{R}^d, \\ u(T, x) = g(x), & x \in \mathbb{R}^d \end{cases}$$

where $r : [0, T] \times \mathbb{R}^d \to \mathbb{R}$ is a continuous bounded below $\mathcal{L}$ is the infinitesimal generator of the diffusion

$$dX_t = b(t, X_t)\, dt + \sigma(t, X_t)\, dW_t$$

given by

$$\mathcal{L} = \sum_{i=1}^d b_i(t, x)\partial_{x_i} + \frac{1}{2}\sum_{i,j=1}^d \left(\sigma\sigma^\top\right)_{ij}(t, x)\partial_{x_i x_j}$$

Then the unique solution of the PDE admits the representation

$$u(t,x) = \mathbb{E}\left[\exp\left(-\int_t^T r(s,X_s)\,ds\right) g(X_T) \;\middle|\; X_t = x\right]$$

When $r \equiv 0$, this reduces to

$$u(t,x) = \mathbb{E}\big[g(X_T) \mid X_t = x\big].$$

Theorem 3.4.4 (Martingale Representation Theorem)**.** *Let $(W_t)_{t\geq 0}$ be an k-dimensional Brownian motion and let $(\mathcal{F}_t)_{t\geq 0}$ be its completed natural filtration. Then every continuous local martingale $(M_t)_{t\in[0,T]}$ with respect to this filtration can be uniquely represented as*

$$M_t = M_0 + \int_0^t \phi_s\,dW_s$$

where $(\phi_t)_{t\in[0,T]} \in \mathcal{L}^2_{loc}$.

Theorem 3.4.5. *Consider the Itô's process*

$$dX_t = b(t,X_t)\,dt + \sigma(t,X_t)\,dW_t$$

with sufficiently regular coefficients. If $p(t,x)$ denotes the probability density of X_t, then p satisfies the Fokker–Planck (or Kolmogorov Forward) equation

$$\partial_t p(t,x) = -\sum_{i=1}^d \partial_{x_i}\big(b_i(t,x)p(t,x)\big) + \frac{1}{2}\sum_{i,j=1}^d \partial_{x_i x_j}\Big(\big(\sigma(t,x)\sigma(t,x)^\top\big)_{ij}\,p(t,x)\Big)$$

Proof Sketch. Let φ be a smooth test function with compact support (or decaying sufficiently fast at infinity). From Itô's formula, we have the fundamental identity

$$\frac{d}{dt}\mathbb{E}[\varphi(X_t)] = \mathbb{E}\big[\mathcal{L}\varphi(X_t)\big] \tag{3.5}$$

Now, express both sides of (3.5) using the density $p(t,x)$ of X_t. The left-hand side becomes

$$\frac{d}{dt}\int_{\mathbb{R}^d} \varphi(x)p(t,x)\,dx = \int_{\mathbb{R}^d} \varphi(x)\,\partial_t p(t,x)\,dx. \tag{3.6}$$

The right-hand side is

$$\int_{\mathbb{R}^d} \mathcal{L}\varphi(x)\,p(t,x)\,dx \tag{3.7}$$

To move the derivatives from φ to p, we apply integration by parts on $\mathbb{R}^d$, assuming that all boundary terms vanish (which holds for sufficiently regular p and φ). For the first-order

term,

$$\int_{\mathbb{R}^d} b_i(x)\partial_{x_i}\varphi(x)\; p(x)\; dx = -\int_{\mathbb{R}^d} \varphi(x)\; \partial_{x_i}\big(b_i(x)p(x)\big)\; dx$$

and for the second-order term,

$$\int_{\mathbb{R}^d} a_{ij}(x)\partial_{x_ix_j}\varphi(x)\; p(x)\; dx = \int_{\mathbb{R}^d} \varphi(x)\; \partial_{x_ix_j}\big(a_{ij}(x)p(x)\big)\; dx$$

Notice the absence of a minus sign in the second equality because two integrations by parts are performed.

Substituting these into (3.7) and equating with (3.6), we obtain

$$\int_{\mathbb{R}^d} \varphi(x)\,\partial_t p(t,x)\,dx = \int_{\mathbb{R}^d} \varphi(x)\left[-\sum_{i=1}^{d}\partial_{x_i}\big(b_i(t,x)p(t,x)\big) + \frac{1}{2}\sum_{i,j=1}^{d}\partial_{x_ix_j}\big(a_{ij}(t,x)p(t,x)\big)\right]dx$$

Since this holds for every test function φ, the integrands must be equal

$$\partial_t p(t,x) = -\sum_{i=1}^{d}\partial_{x_i}\big(b_i(t,x)p(t,x)\big) + \frac{1}{2}\sum_{i,j=1}^{d}\partial_{x_ix_j}\big(a_{ij}(t,x)p(t,x)\big)$$

□

The Fokker-Planck equation was derived rigorously using integration by parts. This technique also introduces the formal adjoint of the infinitesimal generator $\mathcal{L}$, which plays a central role in the theory of stochastic processes and Malliavin calculus.

The formal adjoint $\mathcal{L}^*$ of the generator $\mathcal{L}$ is defined by the relation

$$\int_{\mathbb{R}^d} (\mathcal{L}\varphi)(x)\, p(t,x)\, dx = \int_{\mathbb{R}^d} \varphi(x)\, (\mathcal{L}^* p)(t,x)\, dx$$

which holds for all compactly supported smooth functions φ. From the integration by parts above, we identify

$$\mathcal{L}^* p = -\sum_{i=1}^{d}\partial_{x_i}(bp) + \frac{1}{2}\sum_{i,j=1}^{d}\partial_{x_ix_j}(a_{ij}p)$$

Thus, $\partial_t p = \mathcal{L}^* p$.

Remark 3.4.6. *The formal adjoints of the basic spatial differential operators in $\mathbb{R}^d$ are*

$$(\partial_{x_i})^* = -\partial_{x_i} \qquad \textit{and} \qquad (\partial_{x_ix_j})^* = \partial_{x_ix_j}$$

with appropriate boundary conditions. For the time derivative, we also have $\partial_t^* = -\partial_t$. This simple observation is the starting point for the Malliavin calculus, where the integration by parts technique is extended to the infinite-dimensional Wiener space.

4 Functional Ito Calculus: The Basics

4.1 Path space

For each $t \in [0, \infty)$, let $D([0,t], \mathbb{R}^d)$ denote the set of bounded càdlàg (right-continuous with left limits) $\omega : [0,t] \to \mathbb{R}^d$, and let $C([0,t], \mathbb{R}^d) \subset D([0,t], \mathbb{R}^d)$ denote the subset of continuous paths. We define

$$\Lambda = \bigcup_{t \geq 0} D([0,t], \mathbb{R}^d)$$

We denote an element $\omega \in \Lambda$ by ω_t to indicate that the domain of ω is $[0,t]$, that is, $\omega_t \in D([0,t], \mathbb{R}^d)$. For $r \geq t$, define the flat extension of ω_t to time r as the càdlàg $\omega_{t,r} \in D([0,r], \mathbb{R}^d)$ given by

$$\omega_{t,r}(s) = \omega_t(s \wedge t), \qquad s \in [0,r]$$

In other words, $\omega_{t,r}$ agrees with ω_t on $[0,t]$ and remains constant, equal to $\omega_t(t)$, on $[t,r]$. Finally, for $0 \leq s \leq t$, denote the restriction of ω_t to $[0,s]$ as $\omega_t|_s$.

The set Λ is not naturally a vector space, since there is no natural notion of addition for two paths with different domains. However, it can be endowed with a metric, making it a metric space.

Proposition 4.1.1. *The function $d_\Lambda : \Lambda \times \Lambda \to \mathbb{R}$ given by*

$$d_\Lambda(\omega_s, \nu_t) = d_\infty(\omega_{s,s\vee t}, \nu_{t,s\vee t}) + |s - t|$$

is a metric on Λ. Consequently, (Λ, d_Λ) is a metric space.

Remark 4.1.2. *It is clear that $\omega_{s,s\vee t} = \omega_s$ or $\nu_{t,s\vee t} = \nu_t$ in the definition of d_Λ.*

Proof. It is clear that $d_\Lambda(\omega_s, \omega_s) = 0$ and that d_Λ is a non-negative, symmetric real-valued function. Moreover, for $\omega_s, \nu_s \in \Lambda$ and $t \geq s$, we have $d_\infty(\omega_{s,s\vee t}, \nu_{s,s\vee t}) = d_\infty(\omega_s, \nu_s)$

since $\|\omega_{s,s\vee t}(x) - \nu_{s,s\vee t}(x)\| = \|\omega_s(s) - \nu_s(s)\|$ if $x \in [s,t]$. Then

$$\begin{aligned} d_\infty(\omega_{s,s\vee u}, \mu_{u,s\vee u}) &= d_\infty(\omega_{s,s\vee t\vee u}, \mu_{u,s\vee t\vee u}) \\ &\leq d_\infty(\omega_{s,s\vee t\vee u}, \nu_{t,s\vee t\vee u}) + d_\infty(\nu_{t,s\vee t\vee u}, \mu_{u,s\vee t\vee u}) \\ &= d_\infty(\omega_{s,s\vee t}, \nu_{t,s\vee t}) + d_\infty(\nu_{t,t\vee u}, \mu_{u,t\vee u}) \\ &= d_\infty(\omega_{s,s\vee t}, \nu_{t,s\vee t}) + d_\infty(\nu_{t,t\vee u}, \mu_{u,t\vee u}) \end{aligned}$$

Hence, the triangle inequality follows since we have $|s-u| \leq |s-t| + |t-u|$. Finally, if $d_\Lambda(\omega_s, \nu_t) = 0$ then $d_\infty(\omega_{s,s\vee t}, \nu_{t,s\vee t}) = |s-t| = 0$ which implies $s = t$ and consequently

$$\omega_{s,s\vee t} = \omega_s \text{ and } \nu_{t,s\vee t} = \nu_t \Rightarrow d_\infty(\omega_s, \nu_t) = 0 \Rightarrow \omega_s = \nu_t$$

Thus, d_Λ is a metric on Λ as desired. □

Note that $(D([0,t],\mathbb{R}^d), d_\infty)$ is a subspace of (Λ, d_Λ) for all $t \in [0,\infty)$. Therefore we obtain, for instance, (Λ, d_Λ) is not separable since $(D([0,t],\mathbb{R}^d), d_\infty)$ is not. Another metric question about this space is completeness. But before proving this we first establish an auxiliary and intuitive lemma.

Lemma 4.1.3. *A Cauchy sequence $(\omega_{t_n})_n$ on Λ converges to $\omega_t \in \Lambda$ if and only if $t_n \to t$ and $\omega_t^{(n)} \to \omega_t$ with*

$$\omega_t^{(n)}(s) = \omega_{t_n}(s \wedge t) = \begin{cases} \omega_{t_n}|_t(s) & \text{if } t \leq t_n \\ \omega_{t_n,t}(s) & \text{if } t \geq t_n \end{cases}$$

Proof. First, note that

$$d_\infty(\omega_t^{(n)}, \omega_t) \leq d_\Lambda(\omega_{t_n}, \omega_t) \leq \max\left\{ d_\infty(\omega_t^{(n)}, \omega_t), \sup_{t<s\leq t_n} \|\omega_{t_n}(s) - \omega_t(t)\| \right\} + |t_n - t|$$

Indeed, if $t_n \leq t$ we have $d_\infty(\omega_{t_n,t_n\vee t}, \omega_{t,t_n\vee t}) = d_\infty(\omega_t^{(n)}, \omega_t)$ and if $t \leq t_n$ we have $d_\infty(\omega_{t_n,t_n\vee t}, \omega_{t,t_n\vee t}) = \max\left\{ d_\infty(\omega_t^{(n)}, \omega_t), \sup_{t<s\leq t_n} \|\omega_{t_n}(s) - \omega_t(t)\| \right\}$. Therefore, it follows from $d_\Lambda(\omega_{t_n}, \omega_t) = d_\infty(\omega_{t_n,t_n\vee t}, \omega_{t,t_n\vee t}) + |t_n - t|$.

Now, if $\omega_{t_n} \to \omega_t$, then using the inequality above we have $\omega_t^{(n)} \to \omega_t$ and moreover by the definition of d_Λ in $d_\Lambda(\omega_{t_n}, \omega_t)$ we have $t_n \to t$. Conversely, if $\omega_t^{(n)} \to \omega_t$ and $t_n \to t$, by the inequality above it remains only to prove that $\sup_{t<s\leq t_n} \|\omega_{t_n}(s) - \omega_t(t)\| \to 0$. Since $(\omega_{t_n})_n$ is a Cauchy sequence, for $\varepsilon > 0$ we have for m, n sufficiently large that $d_\Lambda(\omega_{t_m}, \omega_{t_n}) < \frac{\varepsilon}{3}$. Assume, without loss of generality, that $t_m < t_n$. Then by the definition

of d_Λ we have

$$\max\left\{\sup_{t\leq s\leq t_m}\|\omega_{t_n}(s)-\omega_{t_m}(s)\|,\sup_{t_m<s\leq t_n}\|\omega_{t_n}(s)-\omega_{t_m}(t_m)\|\right\}+|t_n-t_m|<\frac{\varepsilon}{3} \quad (4.1)$$

Now, fix n sufficiently large. Since ω_{t_n} is a càdlàg and $t_m\to t$ we have for m sufficiently large that

$$\|\omega_{t_n}(t_m)-\omega_t(t)\|<\frac{\varepsilon}{3} \quad (4.2)$$

Therefore, for $s\in(t,t_n]$ and for m sufficiently large such that $s\in(t_m,t_n]$ we have

$$\|\omega_{t_n}(s)-\omega_t(t)\|\leq\|\omega_{t_n}(s)-\omega_{t_m}(t_m)\|+\|\omega_{t_m}(t_m)-\omega_{t_n}(t_m)\|+\|\omega_{t_n}(t_m)-\omega_t(t)\|$$

The first two terms are each less than $\frac{\varepsilon}{3}$ by (4.1) and the third term is also less than $\frac{\varepsilon}{3}$ by (4.2). Hence, $\|\omega_{t_n}(s)-\omega_t(t)\|<\varepsilon$. Taking the supremum over all $s\in(t,t_n]$ we have $\sup_{t<s\leq t_n}\|\omega_{t_n}(s)-\omega_t(t)\|<\varepsilon$ and making $\varepsilon\to 0^+$ we get that this supremum goes to 0 as we desired. □

Proposition 4.1.4. *The metric space (Λ,d_Λ) is complete.*

Proof. Let $(\omega_{t_n})_n$ be a Cauchy sequence on Λ. So by the definition of d_Λ in $d_\Lambda(\omega_{t_m},\omega_{t_n})$, we have that $(t_n)_n$ is a Cauchy sequence on $[0,T]$ and therefore converges to a point $t\in[0,T]$. Consider the sequence $(\omega_t^{(n)})_n$ on Λ such as in the Lemma (4.1.3). Then, it is easier to see than to write that $d_\infty(\omega_t^{(m)},\omega_t^{(n)})\leq d_\infty(\omega_{t_m,t_m\vee t_n},\omega_{t_n,t_m\vee t_n}),\forall m,n\in\mathbb{N}$, one only needs to suppose, without loss of generality, that $t_m\leq t_n$ and analyze the three cases $t\leq t_m$, $t_m\leq t\leq t_n$ and $t_n\leq t$. So by the fact that $(\omega_{t_n})_n$ is a Cauchy sequence and from the definition of d_Λ it follows that $(\omega_t^{(n)})_n$ is a Cauchy sequence on $D([0,t],\mathbb{R}^d)$. By completeness of $D([0,t],\mathbb{R}^d)$, there is a function $\omega_t\in\Lambda$ such that $\omega_t^{(n)}\to\omega_t$. By Lemma 4.1.3, we have that $\omega_{t_n}\to\omega_t$. This finishes the proof. □

Lemma (4.1.3) is not useful in practice since we need to suppose that the sequence $(\omega_{t_n})_n$ is a Cauchy sequence and the work to prove this fact is as difficult as proving that the sequence converges to some càdlàg known. In more general cases we will prove convergence directly from the definition of d_Λ as in the following lemma.

Proposition 4.1.5. *Let $\omega_t\in\Lambda$ be continuous. Given a sequence $(s_n)_n$ on $[0,t]$ such that s_n converges for some $s\in[0,t]$. Then $\omega_t|_{s_n}\to\omega_t|_s$.*

Proof. Since $\omega_t|_{s_n}(r)=\omega_t|_s(r)=\omega_t(r),\forall r\in[0,s_n\wedge s]$ we have

$$d_\Lambda(\omega_t|_{s_n},\omega_t|_s)=\sup_{s_n\wedge s<r\leq s_n\vee s}\|\omega_t(r)-\omega_t(s_n\wedge s)\|+|s_n-s|$$

As ω_t is continuous and in particular is continuous in s, for $\varepsilon > 0$, we have that $\|\omega_t(r_1) - \omega_t(r_2)\| < \varepsilon$ for all r_1 and r_2 in a sufficiently small interval that contais s. Since $s \in [s_n \wedge s, s_n \vee s]$ and this interval is sufficiently small for n sufficiently large, we have

$$\sup_{s_n \wedge s < r \leq s_n \vee s} \|\omega_t(r) - \omega_t(s_n \wedge s)\| < \varepsilon$$

Letting $\varepsilon \to 0^+$, we obtain this supremum goes to 0 and it is clear that $|s_n \to s|$ goes to 0 too. So $\omega_t|_{s_n} \to \omega_t|_s$. □

Proposition 4.1.5 could be false if ω_t is not continuous. Indeed, let $\omega_2|_1 = \mathbb{1}_{[1,2]}$ and $s_n = 1 - \frac{1}{n}$. Then, $s_n \to 1$ and $d_\Lambda(\omega_2|_{s_n}, \omega_2|_1) = 1 + \frac{1}{n} \to 1$. However, we can get a version of Proposition 4.1.5 for any càdlàg $\omega_t \in \Lambda$.

Proposition 4.1.6. *Let $\omega_t \in \Lambda$. Given a sequence $(s_n)_n$ on $[0, t]$ such that $s_n \downarrow s$. Then $\omega_t|_{s_n} \to \omega_t|_s$.*

Proof. We have

$$d_\Lambda(\omega_t|_{s_n}, \omega_t|_s) = \sup_{s \leq r \leq s_n} \|\omega_t(r) - \omega_t(s)\| + |s_n - s|$$

Since ω_t is right-continuous and $s_n \downarrow s$, we obtain $\omega_t|_{s_n} \to \omega_t|_s$. □

Another important result about the convergence of sequences in Λ is provided below.

Proposition 4.1.7. *Let ω_{t_n} be a sequence on Λ such that ω_{t_n} converges to some ω_t. Then, $\|\omega_{t_n}\|_\infty \to \|\omega_t\|_\infty$*

Proof. It suffices to see that $\|\omega_{t_n}\|_\infty = \|\omega_{t_n, t_n \vee t}\|_\infty$ and $\|\omega_t\|_\infty = \|\omega_{t, t_n \vee t}\|_\infty$. Therefore

$$\begin{aligned}
\|\omega_{t_n}\|_\infty - \|\omega_t\|_\infty &= \|\omega_{t_n, t_n \vee t}\|_\infty - \|\omega_{t, t_n \vee t}\|_\infty \\
&\leq \|\omega_{t_n, t_n \vee t} - \omega_{t, t_n \vee t}\|_\infty \\
&= d_\infty(\omega_{t_n, t_n \vee t}, \omega_{t, t_n \vee t}) \\
&\leq d_\Lambda(\omega_{t_n}, \omega_t)
\end{aligned}$$

Then, $d_\Lambda(\omega_{t_n}, \omega_t) \to 0 \Rightarrow \|\omega_{t_n}\|_\infty \to \|\omega_t\|_\infty$. □

4.2 Path-dependent functionals

A functional is a function $F : \Lambda \to \mathbb{R}^m$. With the distance notion on Λ it is possible talk about continuity of functionals. Precisely, a functional F is continuous on $\omega_s \in \Lambda$ if for each $\varepsilon > 0$ there is a $\delta > 0$ such that $\|F(\omega_s) - F(\nu_t)\| < \varepsilon$ whenever $d_\Lambda(\omega_s, \nu_t) < \delta$. The functional F is said to be continuous if it is continuous in all $\omega_s \in \Lambda$. Finally, F is

uniformly continuous if for each $\varepsilon > 0$ there is a $\delta > 0$ such that $\|F(\omega_s) - F(\nu_t)\| < \varepsilon$ whenever $d_\Lambda(\omega_s, \nu_t) < \delta$.

Example 4.2.1. *The functional $F : \Lambda \to \mathbb{R}^d$ given by $F(\omega_t) = \omega_t(t)$ is uniformly continuous. Indeed, it suffices to note that $\|\omega_s(s) - \nu_t(t)\| \leq d_\infty(\omega_{s,s\vee t}, \nu_{t,s\vee t}) = d_\Lambda(\omega_s, \nu_t)$, i.e., F is Lipschitz continuous and therefore uniformly continuous. Since for $i = 1, ..., d$ we have $|\omega_s(s)_i - \nu_t(t)_i| \leq \|\omega_s(s) - \nu_t(t)\|$ and $|\|\omega_s(s)\| - \|\nu_t(t)\|| \leq \|\omega_s(s) - \nu_t(t)\|$ the functionals $\omega_t \in \Lambda \mapsto \omega_t(t)_i \in \mathbb{R}$ and $\omega_t \in \Lambda \mapsto \|\omega_t(t)\| \in \mathbb{R}$ are also uniformly continuous. By definition of d_Λ, it is immediate that $F(\omega_t) = t$ is uniformly continuous. Finally, the functional $F : \Lambda \to \mathbb{R}$ given by $F(\omega_t) = \|\omega_t\|_\infty$ is uniformly continuous by Proposition 4.1.7.*

Example 4.2.2. *For a non example of continuity, let $T > 0$ and let $F : \Lambda \to \mathbb{R}$ be $F(\omega_t) = \mathbb{1}_{\{T\}}(t)$. If ω_{t_n} is a sequence on Λ converging to $\omega_T \in \Lambda$ with $t_n \neq T, \forall n \in \mathbb{N}$, we have $t_n \to T$ and then $\mathbb{1}_{\{T\}}(t_n) = 0 \neq 1 = \mathbb{1}_{\{T\}}(T)$. So, $F(\omega_{t_n}) \not\to F(\omega_T)$.*

Proposition 4.2.3. *Let $\omega_t \in \Lambda$ be continuous and let $F : \Lambda \to \mathbb{R}^m$ be continuous. Then, $\nu_t : [0, t] \to \mathbb{R}^m$ given by*

$$\nu_t(s) = F(\omega_t|_s), \qquad s \in [0, t]$$

is continuous.

Proof. If $s_n \to s$ in $[0, t]$, Proposition 4.1.5 give us $\omega_t|_{s_n} \to \omega_t|_s$ and continuity of F implies that $F(\omega_t|_{s_n}) \to F(\omega_t|_s)$. The continuity of ν_t follows by composition. □

Proposition 4.2.3 can be reformulated for any càdlàg as follows.

Proposition 4.2.4. *Let $\omega_t \in \Lambda$ and let $F : \Lambda \to \mathbb{R}^m$ be continuous. Then, $\nu_t : [0, t] \to \mathbb{R}^m$ given by*

$$\nu_t(s) = F(\omega_t|_s), \qquad s \in [0, t]$$

is right-continuous.

Proof. The proof is similar to Proposition 4.2.3 but considering $s_n \downarrow s$ in $[0, t]$ and applying Proposition 4.1.6 instead Proposition 4.1.5. □

We give below some simple examples of relevant continuous functionals.

Example 4.2.5. *If $f : \mathbb{R}^{1+d} \to \mathbb{R}^m$ is a continuous function, let $F : \Lambda \to \mathbb{R}^m$ be*

$$F(\omega_t) = f(\omega_t(t))$$

Continuity of F follows from composition of the projection $(t, \omega_t) \mapsto \omega_t$ with the functional $\omega_t \mapsto \omega_t(t)$ (by Example 4.2.1) and f. Actually, it is a extension of Example 4.2.1)

Example 4.2.6. *Let $F : \Lambda \to \mathbb{R}^d$ be*

$$F(\omega_t) = \int_0^t \omega_t(s)\, ds$$

If $\omega_{t_n} \to \omega_t$, we have by Example 4.2.1 that $\omega_{t_n}(t_n) \to \omega_t(t)$ and $\|\omega_{t_n}\|_\infty \to \|\omega_t\|_\infty$. Hence, if $t_n \leq t$, then

$$\begin{aligned}
\|F(\omega_{t_n}) - F(\omega_t)\| &= \left\| \int_0^{t_n} \omega_{t_n}(s)\, ds - \int_0^t \omega_t(s)\, ds \right\| \\
&= \left\| \int_0^{t_n} \omega_{t_n}(s) - \omega_t(s)\, ds + \int_{t_n}^t \omega_{t_n}(t_n) - \omega_t(s)\, ds - \omega_{t_n}(t_n)(t - t_n) \right\| \\
&\leq \int_0^{t_n} \|\omega_{t_n}(s) - \omega_t(s)\|\, ds + \int_{t_n}^t \|\omega_{t_n}(t_n) - \omega_t(s)\|\, ds + \|\omega_{t_n}(t_n)\|(t - t_n) \\
&\leq \int_0^{t_n} d_\Lambda(\omega_{t_n}, \omega_t)\, ds + \int_{t_n}^t d_\Lambda(\omega_{t_n}, \omega_t)\, ds + \|\omega_{t_n}(t_n)\|(t - t_n) \\
&= (d_\Lambda(\omega_{t_n}, \omega_t) + \|\omega_{t_n}(t_n)\|)(t - t_n)
\end{aligned}$$

and if $t_n \geq t$, then

$$\begin{aligned}
\|F(\omega_{t_n}) - F(\omega_t)\| &= \left\| \int_0^{t_n} \omega_{t_n}(s)\, ds - \int_0^t \omega_t(s)\, ds \right\| \\
&= \left\| \int_0^t \omega_{t_n}(s) - \omega_t(s)\, ds + \int_t^{t_n} \omega_{t_n}(s)\, ds \right\| \\
&\leq \int_0^t \|\omega_{t_n}(s) - \omega_t(s)\|\, ds + \int_t^{t_n} \|\omega_{t_n}(s)\|\, ds \\
&\leq \int_0^t d_\Lambda(\omega_{t_n}, \omega_t)\, ds + \|\omega_{t_n}\|_\infty (t_n - t) \\
&= d_\Lambda(\omega_{t_n}, \omega_t) t + \|\omega_{t_n}\|_\infty (t_n - t)
\end{aligned}$$

In both cases, we have that $F(\omega_{t_n}) \to F(\omega_t)$ and therefore F is continuous.

Example 4.2.7. *For $d = 1$, let $F : \Lambda \to \mathbb{R}$ be*

$$F(\omega_t) = \sup_{0 \leq s \leq t} \omega_t(s)$$

Let ω_{t_n} be a sequence on Λ converging to ω_t, Note that $\sup_{0 \leq s \leq t_n} \omega_{t_n}(s) = \sup_{0 \leq s \leq t_n \vee t} \omega_{t_n, t_n \vee t}(s)$

and $\sup_{0\le s\le t}\omega_t(s) = \sup_{0\le s\le t_n\vee t}\omega_{t,t_n\vee t}(s)$. *Then*

$$\begin{aligned}\left|\sup_{0\le s\le t_n}\omega_{t_n}(s) - \sup_{0\le s\le t}\omega_t(s)\right| &= \left|\sup_{0\le s\le t_n\vee t}\omega_{t_n,t_n\vee t}(s) - \sup_{0\le s\le t_n\vee t}\omega_{t,t_n\vee t}(s)\right| \\ &\le \sup_{0\le s\le t_n\vee t}\|\omega_{t_n,t_n\vee t}(s) - \omega_{t,t_n\vee t}(s)\| \\ &\le \sup_{0\le s\le t_n\vee t} d_\infty(\omega_{t_n,t_n\vee t}, \omega_{t,t_n\vee t}) \\ &= d_\infty(\omega_{t_n,t_n\vee t}, \omega_{t,t_n\vee t}) \\ &\le d_\Lambda(\omega_{t_n}, \omega_t)\end{aligned}$$

Hence, F is Lipschitz continuous and therefore uniformly continuous.

The following lemma shows that continuity along a continuous path is a form of uniform continuity with respect to all of its restricions. This result will be useful in the proof of Functional Ito Formula.

Lemma 4.2.8 (Λ-Lemma)**.** *Let $\omega_T \in \Lambda$ be continuous, and let $F : \Lambda \to \mathbb{R}^m$ be continuous. Then for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|F(\omega_T|_s) - F(\nu_t)\| < \varepsilon$ whenever $d_\Lambda(\omega_T|_s, \nu_t) < \delta$.*

Proof. Suppose that it is not true. Then, there is $\varepsilon > 0$ and sequences $(s_n)_n$ and $(t_n)_n$ on $[0,T]$ such that $d_\Lambda(\omega_T|_{s_n}, \nu_{t_n}) \to 0$ but $\|F(\omega_T|_{s_n}) - F(\nu_{t_n})\| \ge \varepsilon$. By Bolzano-Weierstrass's Theorem, we have a subsequence $(s_{n_k})_k$ of $(s_n)_n$ which converges for some $s \in [0,T]$. By Proposition 4.1.5, we have $\omega_T|_{s_{n_k}} \to \omega_T|_s$ and then continuity of functional F give us that $F(\omega_T|_{s_{n_k}}) \to F(\omega_T|_s)$. Furthermore, $\omega_T|_{s_{n_k}} \to \omega_T|_s$ implies $\nu_{t_{n_k}} \to \omega_T|_s$ since $d_\Lambda(\omega_T|_{s_n}, \nu_{t_n}) \to 0$. Hence, $F(\nu_{t_n}) \to F(\omega_T|_s)$. Consequently, taking the limit in $\|F(\omega_T|_{s_{n_k}}) - F(\nu_{t_{n_k}})\| \ge \varepsilon$ when $n \to \infty$ we get $0 \ge \varepsilon$ which is a contradiction. □

Lemma 4.2.8 could be false if ω_T is not continuous. To see this, let $\omega_2 = \mathbb{1}_{[1,2]}$, $F(\omega_t) = \omega_t(t)$ and $t_n = 1 - \frac{1}{n}$. Then, $t_n \to 1$ and by Proposition 4.1.5 $\omega_2|_{t_n} \to \omega_2|_1$. However, $F(\omega_2|_1) - F(\omega_2|_{t_n}) = 1, \forall n \in \mathbb{N}$.

To finish this section, we can consider stochastic processes of the form $(F(X_t))_{t\ge 0}$. To avoid confusion with the classic stochastic theory where $X_t \in \mathbb{R}^d$, we adopt the convention that we have a stochastic process $(x_t)_{t\ge 0}$ with $x_t \in \mathbb{R}^d$ and X_t represents an element of $D([0,t],\mathbb{R}^d)$ such that x_t its terminal value, i.e., $x_t = X_t(t)$. Since we want $X_t \in D([0,t],\mathbb{R}^d)$ we assume that $(x_t)_{t\ge 0}$ is a càdlàg stochastic process, i.e., $t \mapsto x_t$ is a càdlàg. Actually, if we suppose that $(x_t)_{t\ge 0}$ is continuous and then $X_t \in C([0,t],\mathbb{R}^d)$, for all $t \in [0,\infty)$.

Remark 4.2.9. *Notice that for $t \le T$ we have $X_t = X_T|_t$.*

4.3 Horizontal and vertical derivatives

To formulate a differential functional calculus on Λ we need to specify how the path is perturbed.

The first perturbation is the flat extension, one that has already been seen before. Remember that the flat extension of ω_t to time $r \geq 0$ for $r \geq t$ is the path $\omega_{t,r} \in D([0,r],\mathbb{R}^d)$ given that

$$\omega_{t,r}(s) = \omega_t(s \wedge t), \qquad s \in [0,r]$$

In other words, $\omega_{t,r}$ agrees with ω_t on $[0,t]$ and remains constant, equal to $\omega_t(t)$ on $[t,r]$. This corresponds to make the time frozen from t until r.

The second perturbation is the vertical shift by $r \in \mathbb{R}^d$, which for ω_t consists of the path $\omega_t^r \in D([0,t],\mathbb{R}^d)$ given by

$$\omega_t^r(s) = \omega_t(s) + r\mathbb{1}_{\{t\}}(s), \qquad s \in [0,t]$$

In other words, ω_t^r agrees with ω_t on $[0,t)$ and change from $\omega_t(t)$ to $\omega_t(t)+r$ in t. This corresponds to jump at time t.

Proposition 4.3.1. *Let $F : \Lambda \to \mathbb{R}^m$ be a continuous functional and $\omega_t \in \Lambda$. Then the functions $f : [0,\infty) \to \mathbb{R}^m$, $g : \mathbb{R}^d \to \mathbb{R}^m$ given by $f(h) = F(\omega_{t,t+h})$ and $g(h) = F(\omega_t^h)$ are continuous.*

Proof. If $h_n \in [0,\infty) \to h \in [0,\infty)$, then $d_\Lambda(\omega_{t,t+h_n},\omega_{t,t+h}) \to 0$ because it is imeddiate that $d_\infty(\omega_{t,(t+h_n)\vee(t+h)},\omega_{t,(t+h_n)\vee(t+h)}) = 0$ and then $d_\Lambda(\omega_{t,t+h_n},\omega_{t,t+h}) = |h_n - h| \to 0$. Now, if $h_n \in \mathbb{R}^d \to h \in \mathbb{R}^d$, then $d_\Lambda(\omega_t^{h_n},\omega_t^h) = d_\infty(\omega_t^{h_n},\omega_t^h) = \|h_n - h\| \to 0$. Hence, f and g are continuous. Actually, f and g are uniformly continuous and the proof can be adapted changing $(h_n)_n$ with $h_n \to h$ by $(h_n)_n$ and $(r_n)_n$ with $h_n - r_n \to 0$. □

With this perturbations, we define for a functional $F : \Lambda \to \mathbb{R}^m$ the horizontal derivative in ω_t by the limit

$$\Delta_t F(\omega_t) = \lim_{h \downarrow 0} \frac{F(\omega_{t,h}) - F(\omega_t)}{h}$$

and the vertical derivative in ω_t being a linear operator $\Delta_x F(\omega_t) : \mathbb{R}^d \to \mathbb{R}^m$ such that

$$F(\omega_t^h) = F(\omega_t) + \Delta_x F(\omega_t) \cdot h + o(\|h\|)$$

A similar result to Proposition 4.3.1 is described below.

Proposition 4.3.2. *Let $F : \Lambda \to \mathbb{R}^m$ be a functional with horizontal and vertical derivatives and $\omega_t \in \Lambda$. Then the function $f : [0,\infty) \to \mathbb{R}^m$ given by $f(h) = F(\omega_{t,t+h})$ is*

right-differentiable and the function $g : \mathbb{R}^d \to \mathbb{R}^m$ given by $g(h) = F(\omega_t^h)$ is differentiable.

Proof. The differentiability of f is given by

$$\lim_{h\downarrow 0} \frac{f(r+h) - f(r)}{h} = \lim_{h\downarrow 0} \frac{F(\omega_{t,t+r+h}) - F(\omega_{t,t+r})}{h} = \Delta_t F(\omega_{t,t+r})$$

Hence, $f'_+(r) = \Delta_t F(\omega_{t,t+r})$. Now, the differentiability of g is given by

$$g(r+h) = F(\omega_t^{r+h}) = F(\omega_t^r) + \Delta_x F(\omega_t^r) \cdot h + o(\|h\|) = g(r) + \Delta_x F(\omega_t^r) \cdot h + o(\|h\|)$$

from which $g'(r) = \Delta_x F(\omega_t^r)$. □

Proposition 4.3.3. *We have the following properties about the horizontal and vertical derivatives of a functional F:*

1. *if $\Delta_x F(\omega_t)$ exists, then it is indeed the only linear operator that satisfies the definition of vertical derivative.*

2. *If F admits the vertical derivative in ω_t then for $i = 1, ..., d$, the limit*

$$\frac{\partial F}{\partial x_i}(\omega_t) = \lim_{h\to 0} \frac{F(\omega_t^{he_i}) - F(\omega_t)}{h}$$

   exists and it coincides with $\Delta_x F(\omega_t) \cdot e_i$ and will be said the i-th partial derivative in ω_t.

3. *If $F, G : \Lambda \to \mathbb{R}^m$ are functionals with horizontal and vertical derivatives and $\lambda \in \mathbb{R}$, then $F + \lambda G$ has horizontal and vertical derivatives such that*

$$\Delta_t(F + \lambda G) = \Delta_t F + \lambda \Delta_t G \text{ and } \Delta_x(F + \lambda G) = \Delta_x F + \lambda \Delta_x G$$

4. *(Product Rule) If $F : \Lambda \to \mathbb{R}^{m\times n}$ and $G : \Lambda \to \mathbb{R}^{n\times p}$ are functionals with horizontal and vertical derivatives then FG has horizontal and vertical derivatives such that*

$$\Delta_t(FG) = (\Delta_t F)G + F(\Delta_t G) \text{ and } \Delta_x(FG) = (\Delta_x F)G + F(\Delta_x G)$$

5. *(Quotient Rule) If $F : \Lambda \to \mathbb{R}^m$ and $G : \Lambda \to \mathbb{R}$ are functionals with horizontal and vertical derivatives then F/G has horizontal and vertical derivatives such that*

$$\Delta_t(F/G) = \frac{(\Delta_t F)G - F(\Delta_t G)}{G^2} \text{ and } \Delta_x(F/G) = \frac{(\Delta_x F)G - F(\Delta_x G)}{G^2}$$

6. *(Chain Rule): If $F : \Omega \to \mathbb{R}^m$ is a functional with horizontal and vertical derivatives and $f : \mathbb{R}^m \to \mathbb{R}^n$ is a differentiable function, the $f \circ F$ has horizontal and*

vertical derivatives such that

$$\Delta_t(f \circ F)(\omega_t) = f'(F(\omega_t))\Delta_t F(\omega_t) \text{ and } \Delta_x(f \circ F)(\omega_t) = f'(F(\omega_t))\Delta_x F(\omega_t)$$

Proof. Define $f : [0, \infty) \to \mathbb{R}^m$, $g : \mathbb{R}^d \to \mathbb{R}^m$ by $f(h) = F(\omega_{t,t+h})$ and $g(h) = F(\omega_t^h)$. Then, by Proposition 4.3.2, we have that f is right-differentiable and g is differentiable with $f'_+(r) = \Delta_t F(\omega_{t,t+r})$ and $g'(r) = \Delta_x F(\omega_t^r)$. In particular, $f'_+(0) = \Delta_t F(\omega_t)$ and $g'(0) = \Delta_x F(\omega_t)$. The properties follows from the classical differential calculus. For instance, the product rule will be

$$\Delta_t(FG)(\omega_t) = (f_1 g_1)'(0) = f_1'(0)g_1(0) + f_1(0)g_1'(0) = (\Delta_t(F))G + F(\Delta_t G)$$

and

$$\Delta_x(FG)(\omega_t) = (f_2 g_2)'(0) = f_2'(0)g_2(0) + f_2(0)g_2'(0) = (\Delta_x(F))G + F(\Delta_x G)$$

where $f_1 : [0, \infty) \to \mathbb{R}^{m\times n}$, $g_1 : [0, \infty) \to \mathbb{R}^{n\times p}$, $f_2 : \mathbb{R}^d \to \mathbb{R}^{m\times n}$, $g_2 : \mathbb{R}^d \to \mathbb{R}^{n\times p}$ are given by $f_1(h) = F(\omega_{t,t+h})$, $g_1(h) = G(\omega_{t,t+h})$, $f_2(h) = F(\omega_t^h)$, $g_2(h) = G(\omega_t^h)$. □

Remark 4.3.4. *By property 2 in Proposition 4.3.3 we can assume that $\Delta_x F(\omega_t)$ is a matrix $m \times d$ with i-th column being $\Delta_x F(\omega_t) \cdot e_i = \frac{\partial F}{\partial x_i}(\omega_t)$.*

We analyze some functionals have been seen in the last section if they have or no horizontal and vertical derivatives.

Example 4.3.5. *For $F(\omega_t) = f(t, \omega_t(t))$, where $f : \mathbb{R}^{1+d} \to \mathbb{R}^m$ is differentiable, we have*

$$\begin{aligned}\Delta_t F(\omega_t) &= \lim_{h\downarrow 0} \frac{f(t+h, \omega_{t,t+h}(t+h)) - f(t, \omega_t(t))}{h} \\ &= \lim_{h\downarrow 0} \frac{f(t+h, \omega_t(t)) - f(t, \omega_t(t))}{h} = \frac{\partial f}{\partial t}(t, \omega_t(t))\end{aligned}$$

and

$$\begin{aligned}F(\omega_t^h) &= f(t, \omega_t^h(t)) = f(t, \omega_t(t)) + \partial_x f(t, \omega_t(t)) \cdot h + o(\|h\|) \\ &= F(\omega_t) + \partial_x f(t, \omega_t(t)) \cdot h + o(\|h\|)\end{aligned}$$

from wich we have that $\Delta_x F(\omega_t) = \partial_x f(t, \omega_t(t))$.

Example 4.3.6. *For $F(\omega_t) = \int_0^t \omega_t(s)\, ds$, we have*

$$\Delta_t F(\omega_t) = \lim_{h\downarrow 0} \frac{\int_0^{t+h} \omega_{t+h}(s)\, ds - \int_0^t \omega_t(s)\, ds}{h} = \lim_{h\downarrow 0} \frac{\int_t^{t+h} \omega_t(t)\, ds}{h} = \omega_t(t)$$

and we see that $F(\omega_t^h) = \int_0^t \omega_t^h(s)\,ds = \int_0^t \omega_t(s)\,ds = F(\omega_t)$ *since a perturbation in terminal value do not change the Lebesgue integral. Hence,* $\Delta_x F(\omega_t) = 0$.

Example 4.3.7. *For* $d = 1$ *and* $F(\omega_t) = \sup_{0\leq s\leq t} \omega_t(s)$, *we have*

$$\Delta_t F(\omega_t) = \lim_{h\downarrow 0} \frac{F(\omega_{t,t+h}) - F(\omega_t)}{h} = \lim_{h\downarrow 0} \frac{\sup_{0\leq s\leq t+h} \omega_{t+h}(s) - \sup_{0\leq s\leq t} \omega_t(s)}{h} = 0$$

since $\sup_{0\leq s\leq t+h} \omega_{t+h}(s) = \sup_{0\leq s\leq t} \omega_t(s)$. *However, the existence of* $\Delta_x F$ *depends of* ω_t. *Indeed, we have three cases to analyze:*

- *if* $\omega_t(t) < F(\omega_t)$, *then for a sufficiently small* $h \in \mathbb{R}$, *we get* $\omega_t(t)+h < \sup_{0\leq s\leq t} \omega_t(s)$, *i.e.,* $F(\omega_t^h) = F(\omega_t)$ *and then* $\Delta_x F(\omega_t) = 0$.

- *if* $\omega_t(t) = F(\omega_t)$ *and* $\omega_t(t) > \sup_{0\leq s<t} \omega_t(s)$, *then for a sufficiently small* $h \in \mathbb{R}$, *we get* $\omega_t(t) + h > \sup_{0\leq s\leq t} \omega_t(s)$, *i.e.,* $F(\omega_t^h) = \omega_t(t) + h = F(\omega_t) + 1 \cdot h$ *and then* $\Delta_x F(\omega_t) = 1$.

- *if* $\omega_t(t) = F(\omega_t)$ *and* $\omega_t(t) = \sup_{0\leq s<t} \omega_t(s)$, *then for* $h \geq 0$ *we get* $\omega_t(t) + h \geq \sup_{0\leq s\leq t} \omega_t(s)$ *and for* $h < 0$ *we get* $\omega_t(t) + h < \sup_{0\leq s\leq t} \omega_t(s)$. *Hence, if* $h \geq 0$, *we have* $F(\omega_t^h) = F(\omega_t)+h$ *and if* $h < 0$ *we have* $F(\omega_t^h) = F(\omega_t)$. *Consequently,* $\Delta_x F(\omega_t)$ *do not exists.*

Therefore, we obtain

$$\Delta_x F(\omega_t) = \begin{cases} 0 & \text{if } \omega_t(t) < F(\omega_t) \\ 1 & \text{if } \omega_t(t) = F(\omega_t) \text{ and } \omega_t(t) > \sup_{0\leq s<t} \omega_t(s) \\ \nexists & \text{if } \omega_t(t) = F(\omega_t) \text{ and } \omega_t(t) = \sup_{0\leq s<t} \omega_t(s) \end{cases}$$

Note that if $F : \Lambda \to \mathbb{R}^m$ is a functional with horizontal and vertical derivatives, then $\Delta_t F : \Lambda \to \mathbb{R}^m$ and $\Delta_x F : \Lambda \to \mathbb{R}^{m\times d}$ are also functionals. In particular, we can define $\Delta_{tt} F : \Lambda \to \mathbb{R}^m$, $\Delta_{xx} F : \Lambda \to \mathbb{R}^{m\times d\times d}$, $\Delta_{tx} F : \Lambda \to \mathbb{R}^{m\times d}$, $\Delta_{xt} F : \Lambda \to \mathbb{R}^{m\times d}$ and higher-order derivatives when they exist. A functional with continuous horizontal and vertical derivatives is said to be a $C^{1,1}$ functional. More generally, a functional with i successive continuous horizontal derivatives and j successive continuous vertical derivatives is said to be a $C^{i,j}$ functional.

4.4 Functional Ito Formula

Before to prove the Functional Itô's Formula, we establish important lemmas that will be used along the proof

Lemma 4.4.1. *Let $T > 0$, $(\pi_n)_n$ be a sequence of partitions of $[0,T]$, $\pi_n = (t_k^{(n)})_{k=0}^{k_n}$, converging to identity and $f : [0,T] \to \mathbb{R}^d$ continuous. For each $n \in \mathbb{N}$, define as the piecewise constant approximation from below of f the funcion $f_n : [0,T] \to \mathbb{R}^d$ given by*

$$f_n(t) = \sum_{k=1}^{k_n} f(t_{k-1}^{(n)}) \mathbb{1}_{[t_{k-1}^{(n)}, t_k^{(n)})}(t) + f(T)\mathbb{1}_{\{T\}}(t), \qquad t \in [0,T]$$

Then, $f_n \to f$, i.e., f_n converges uniformly to f.

Proof. Since f is continuous and defined in a compact set $[0,T]$, it is uniformly continuous. So, for $\varepsilon > 0$, there is $\delta > 0$ such that $\|f(s) - f(t)\| < \varepsilon$ whenever $|s - t| < \delta$.

Since $(\pi_n)_n$ converges to identity, there is $n_0 \in \mathbb{N}$ such that for all $n > n_0$ we have $|t_k^{(n)} - t_{k-1}^{(n)}| < \delta, \forall k = 1, ..., k_n$. Then, $\|f(s) - f(t)\| < \varepsilon, \forall s, t \in [t_{k-1}^{(n)}, t_k^{(n)}]$ for any $n > n_0$ and $k = 1, ..., k_n$.

It is immediate that $f_n(T) \to f(T)$ because $f_n(T) = f(T), \forall n \in \mathbb{N}$. Now, for $t \in [0,T)$, let $n > n_0$ and $k \in \{1, ..., k_n\}$ such that $t \in [t_{k-1}^{(n)}, t_k^{(n)})$. Hence,

$$\|f_n(t) - f(t)\| = \|f(t_{k-1}^{(n)}) - f(t)\| < \varepsilon$$

Taking the supremum over all $t \in [0,T]$, we get that $f_n \to f$. □

Lemma 4.4.2. *Let $T > 0$ and let $x : [0,\infty) \times \Omega \to \mathbb{R}$ be a process with finite quadratic variation, i.e., $\langle x \rangle_T < \infty$. Suppose $f_n, f : [0,T] \times \Omega \to \mathbb{R}^d$ are such that, for each fixed $\omega \in \Omega$,*

- *the mappings $t \mapsto f_n(t,\omega)$ and $t \mapsto f(t,\omega)$ are Borel-measurable;*
- $\sup_{t \in [0,T]} \|f_n(t,\omega) - f(t,\omega)\| \to 0$

Then, for every $\omega \in \Omega$,

$$\int_0^T f_n(t,\omega)\, d\langle x \rangle_t(\omega) \to \int_0^T f(t,\omega)\, d\langle x \rangle_t(\omega).$$

Notably, if $(y_t)_{t \geq 0}$ and $(y_t^{(n)})_{t \geq 0}$ are stochastic processes in $\mathbb{R}^d$ with $\sup_{t \in [0,T]} \|y_t^{(n)} - y_t\| \to 0$. Then,

$$\int_0^T y_t^{(n)}\, d\langle x \rangle_t \to \int_0^T y_t\, d\langle x \rangle_t$$

Proof. Fix $\omega \in \Omega$. Then,

$$\left\| \int_0^T f_n(t,\omega)\, d\langle x\rangle_t - \int_0^T f(t,\omega)\, d\langle x\rangle_t \right\| \leq \int_0^T \|f_n(t,\omega) - f(t,\omega)\|\, d\langle x\rangle_t$$
$$\leq \left(\sup_{t\in[0,T]} \|f_n(t,\omega) - f(t,\omega)\| \right) \langle x\rangle_T(\omega)$$

Since $\langle x\rangle_T(\omega) < \infty$ and the supremum tends to 0, the right-hand side tends to 0. Hence, the desired convergence holds. □

Remark 4.4.3. *In the Lemma 4.4.2, if* $\| \int_0^T f(t,\omega)\, d\langle x\rangle_t(\omega)\| < \infty$*, then we can also prove it by a simple application of the Dominated Convergence Theorem since a uniform convergence implies a pointwise convergence and integrability of* f *implies integrability of* f_n *by the uniform convergence and the finite measure of* $[0,T]$.

Lemma 4.4.4. *Let* $T > 0$ *and let* $x : [0,\infty) \times \Omega \to \mathbb{R}$ *be a continuous semimartingale. Suppose* $(y_t)_{t\geq 0}$ *and* $(y_t^{(n)})_{t\geq 0}$ *are adapted (to filtration induced by* x*) stochastic processes in* $\mathbb{R}^d$ *such that the paths* Y_T *and* $Y_T^{(n)}$ *are right-continuous and suppose also that* $\sup_{t\in[0,T]} \|y_t^{(n)} - y_t\| \to 0$ *and* $\sup_{t\in[0,T]} \|y_t\| < \infty$*. Then,*

$$\sup_{0\leq t\leq T} \left\| \int_0^t y_s^{(n)}\, dx_s - \int_0^t y_s\, dx_s \right\| \xrightarrow{\mathbb{P}} 0$$

In particular,

$$\int_0^t y_s^{(n)}\, dx_s \xrightarrow{\mathbb{P}} \int_0^T y_s\, dx_s$$

Proof. The processes $(y_t)_{t\geq 0}$ and $(y_t^{(n)})_{t\geq 0}$ are progressive by Proposition 3.4 of [6]. The condition $\sup_{t\in[0,T]} \|y_t^{(n)} - y_t\| \to 0$ give us that $y_t^{(n)} \to y_t$, i.e., $y_t^{(n)}$ converges pointwise (and then almost surely) to y_t. Furthermore, this condition together with the condition $\sup_{t\in[0,T]} \|y_t^{(n)}\| < \infty$ implies by a triangular inequality that there is $n_0 \in \mathbb{N}$ such that $\sup_{n>n_0} \sup_{t\in[0,T]} \|y_t^{(n)}\| < \infty$. So, the proof is a simple application of Dominated Convergence Theorem for Stochastic Integrals (Proposition 5.8 of [6]). □

Now, we are able to prove the Functional Ito Formula.

Theorem 4.4.5. *(Functional Ito Formula) Let* $F : \Lambda \to \mathbb{R}$ *a* $C^{1,2}$ *functional and* $(x_t)_{t\geq 0}$ *a continuous semimartingale. Then for* $T > 0$ *we have*

$$F(X_T) = F(X_0) + \int_0^T \Delta_t F(X_t)\, dt + \int_0^T \Delta_x F(X_t)\, dx_t + \frac{1}{2} \int_0^T \text{tr}(\Delta_{xx} F(X_t)\, d\langle x\rangle_t)\ \mathbb{P}\text{- a.s.}$$

Remark 4.4.6. *The integrals appearing in Theorem 4.4.5 are well-defined under the stated assumptions. Since $F \in C^{1,2}$, the functionals $\Delta_t F$, $\Delta_x F$, and $\Delta_{xx} F$ are continuous. For a continuous semimartingale $(x_t)_{t\geq 0}$, the map $t \mapsto X_t$ is continuous by Proposition 4.1.5. Therefore, the compositions $t \mapsto \Delta_t F(X_t)$, $t \mapsto \Delta_x F(X_t)$ and $t \mapsto \Delta_{xx} F(X_t)$ are continuous processes, and hence progressively measurable and satisfies the integrability conditions.*

Proof. Let $(\pi_n)_n$ be a sequence of partitions of $[0,T]$, $\pi_n = (t_k^{(n)})_{k=0}^{k_n}$, converging to identity. For $k = 1, ..., k_n$, we define

$$ {}_nY_{t_k^{(n)}}(s) = \sum_{i=1}^{k} x_{t_{i-1}^{(n)}} \mathbb{1}_{[t_{i-1}^{(n)}, t_i^{(n)})}(s) + x_{t_k^{(n)}} \mathbb{1}_{\{t_k^{(n)}\}}(s), \qquad s \in [0, t_k^{(n)}] $$

In other words, ${}_nY_{t_k^{(n)}}$ is a piecewise constant approximation from below of $X_{t_k^{(n)}}$. Also define

$$ {}_nZ_{t_k^{(n)}}(s) = \begin{cases} {}_nY_{t_k^{(n)}}(s) & \text{if } s < t_k^{(n)} \\ x_{t_{k-1}^{(n)}} & \text{if } s = t_k^{(n)} \end{cases}, \qquad s \in [0, t_k^{(n)}] $$

In other words, ${}_nZ_{t_k^{(n)}}$ is also a piecewise constant approximation from below of $X_{t_k^{(n)}}$, but only on $[0, t_k^{(n)})$, with ${}_nZ_{t_k^{(n)}}(t_k^{(n)}) = x_{t_{k-1}^{(n)}}$, i.e., ${}_nZ_{t_k^{(n)}}$ is continuous at $t_k^{(n)}$. By definitions above, for $k = 1, ..., k_n$, we have

$$ {}_nY_{t_k^{(n)}} =_n Z_{t_k^{(n)}}^{x_{t_k^{(n)}} - x_{t_{k-1}^{(n)}}} \qquad \text{and} \qquad {}_nZ_{t_k^{(n)}} =_n Y_{t_{k-1}^{(n)}, t_k^{(n)}} $$

With this in mind, we obtain the following telescopic sum

$$ \begin{aligned} & F(X_T) - F(X_0) \\ &= F(X_T) - F({}_nY_{t_{k_n}^{(n)}}) + \sum_{k=1}^{k_n} F({}_nY_{t_k^{(n)}}) - F({}_nY_{t_{k-1}^{(n)}}) \\ &= F(X_T) - F({}_nY_{t_{k_n}^{(n)}}) + \sum_{k=1}^{k_n} F({}_nY_{t_k^{(n)}}) - F({}_nZ_{t_k^{(n)}}) + \sum_{k=1}^{k_n} F({}_nZ_{t_k^{(n)}}) - F({}_nY_{t_{k-1}^{(n)}}) \\ &= F(X_T) - F({}_nY_{t_{k_n}^{(n)}}) + \sum_{k=1}^{k_n} F({}_nZ_{t_k^{(n)}}^{x_{t_k^{(n)}} - x_{t_{k-1}^{(n)}}}) - F({}_nZ_{t_k^{(n)}}) + \sum_{k=1}^{k_n} F({}_nY_{t_{k-1}^{(n)}, t_k^{(n)}}) - F({}_nY_{t_{k-1}^{(n)}}) \end{aligned} $$

Define

$$A_n = F(X_T) - F({}_nY_{t_{k_n}^{(n)}})$$
$$B_n = \sum_{k=1}^{k_n} F({}_nZ_{t_k^{(n)}}^{x_{t_k^{(n)}} - x_{t_{k-1}^{(n)}}}) - F({}_nZ_{t_k^{(n)}})$$
$$C_n = \sum_{k=1}^{k_n} F({}_nY_{t_{k-1}^{(n)}, t_k^{(n)}}) - F({}_nY_{t_{k-1}^{(n)}})$$

We then analyze each one of this terms when $n \to \infty$.

- $\lim_{n\to\infty} A_n = 0$:

  ${}_nY_{t_{k_n}^{(n)}}$ is a piecewise constant approximation from below of X_T on $[0,T]$. Then, Lemma 4.4.1 give us ${}_nY_{t_{k_n}^{(n)}} \to X_T$. Continuity of F implies $F({}_nY_{t_{k_n}^{(n)}}) \to F(X_T)$. Hence, $A_n \to 0$. This convergence is pointwise, consequently, is almost surely and moreover in probability.

- $\lim_{n\to\infty} B_n = \int_0^T \Delta_x F(X_t)\, dx_t + \frac{1}{2}\int_0^T \Delta_{xx} F(X_t)\, d\langle x\rangle_t$:

  By Proposition 4.3.1 and Proposition 4.3.2, the function $r \in \mathbb{R}^d \to F({}_nZ^r_{t_k^{(n)}})$ is C^2 with derivative $r \mapsto \Delta_x F({}_nZ^r_{t_k^{(n)}})$ and second-order derivative $r \mapsto \Delta_{xx} F({}_nZ^r_{t_k^{(n)}})$. So, by Itô's Formula, we have

$$F({}_nZ_{t_k^{(n)}}^{x_{t_k^{(n)}} - x_{t_{k-1}^{(n)}}}) - F({}_nZ_{t_k^{(n)}}) = \int_{t_{k-1}^{(n)}}^{t_k^{(n)}} \Delta_x F({}_nZ^{x_r}_{t_k^{(n)}})\, dx_r + \frac{1}{2}\int_{t_{k-1}^{(n)}}^{t_k^{(n)}} \mathrm{tr}(\Delta_{xx} F({}_nZ^{x_r}_{t_k^{(n)}})\, d\langle x\rangle_r)$$

  Let $i_n : [0,T] \to \{1, ..., k_n\}$ be $i_n(s) = k$ if $s \in [t_{k-1}^{(n)}, t_k^{(n)})$ and $i_n(T) = k_n$. Hence,

$$\begin{aligned} B_n &= \sum_{k=1}^{k_n} F({}_nZ_{t_k^{(n)}}^{x_{t_k^{(n)}} - x_{t_{k-1}^{(n)}}}) - F({}_nZ_{t_k^{(n)}}) \\ &= \int_0^T \Delta_x F({}_nZ^{x_r}_{t_{i_n(r)}^{(n)}})\, dx_r + \frac{1}{2}\int_0^T \mathrm{tr}(\Delta_{xx} F({}_nZ^{x_r}_{t_{i_n(r)}^{(n)}})\, d\langle x\rangle_r) \end{aligned}$$

  Notice that ${}_nZ^{x_r}_{t_{i_n(r)}^{(n)}}$ is a piecewise constant approximation from below of $X_{t_{i_n(r)}^{(n)}}$ and then the flat extension of ${}_nZ^{x_r}_{t_{i_n(r)}^{(n)}}$ to time r is a piecewise constant approximation from below of X_r. Since X_r is continuous and defined in a compact set $[0,r]$, Lemma 4.4.1 give us $d_\Lambda({}_nZ^{x_r}_{t_{i_n(r)}^{(n)}}, X_r) \to 0$. Futhermore, it is immediate that $({}_nZ^{x_r}_{t_{i_n(r)}^{(n)}})_{r\in[0,T]}$ is adapted.

  Since $\Delta_{xx}F$ is continuous, we have $\sup_{r\in[0,T]} \|\Delta_{xx} F({}_nZ^{x_r}_{t_{i_n(r)}^{(n)}}) - \Delta_{xx} F(X_r)\| \to 0$ from Λ-Lemma (Lemma 4.2.8). Because $(x_t)_{t\geq 0}$ is a continuous semimartingale,

its quadratic variation $\langle x \rangle_t$ is finite and then Lemma 4.4.2 implies

$$\frac{1}{2}\int_0^T \text{tr}(\Delta_{xx}F({}_nZ^{x_r}_{t^{(n)}_{i_n(r)}})\, d\langle x\rangle_r) \to \frac{1}{2}\int_0^T \text{tr}(\Delta_{xx}F(X_r)\, d\langle x\rangle_r)$$

This convergence is pointwise, consequently, is almost surely and moreover in probability.

Again by the Λ-Lemma, we have $\sup_{r\in[0,T]} \|\Delta_x F({}_nZ^{x_r}_{t^{(n)}_{i_n(r)}}) - \Delta_x F(X_r)\| \to 0$. For ω, $X_T(\omega)$ is continuous and so is $X_r(\omega) = X_T|_r(\omega)$. Since $\Delta_x F$ is continuous, Proposition 4.2.3 implies that the map $r \in [0,T] \mapsto \Delta_x F(X_r) \in \mathbb{R}^m$ is continuous and thus its image is compact, i.e., $\sup_{r\in[0,T]} \|\Delta_x F(X_r)\| < \infty$. Finally, since we have $r \mapsto_n Z^{x_r}_{t^{(n)}_{i_n(r)}} \in \Lambda$, Proposition 4.2.4 implies that $\Delta_x F({}_nZ^{x_r}_{t^{(n)}_{i_n(r)}})$ is right-continuous. Hence, Lemma 4.4.4 tell us that

$$\int_0^T \Delta_x F({}_nZ^{x_r}_{t^{(n)}_{i_n(r)}})\, dx_r \xrightarrow{\mathbb{P}} \int_0^T \Delta_x F(X_r)\, dx_r$$

So, we have

$$\begin{aligned} B_n &= \int_0^T \Delta_x F({}_nZ^{x_r}_{t^{(n)}_{i_n(r)}})\, dx_r + \frac{1}{2}\int_0^T \text{tr}(\Delta_{xx}F({}_nZ^{x_r}_{t^{(n)}_{i_n(r)}})\, d\langle x\rangle_r) \\ &\xrightarrow{\mathbb{P}} \int_0^T \Delta_x F(X_r)\, dx_r + \frac{1}{2}\int_0^T \text{tr}(\Delta_{xx}F(X_r)\, d\langle x\rangle_r) \end{aligned}$$

- $\lim_{n\to\infty} C_n = \int_0^T \Delta_t F(X_t)\, dt$:

By Proposition 4.3.1 and Proposition 4.3.2, the function $r \in [0,\infty) \to F({}_nY_{t^{(n)}_{k-1}, t^{(n)}_{k-1}+r})$ is right-differentiable with right-derivative continuous and this right-derivative is $r \mapsto \Delta_t F({}_nY_{t^{(n)}_{k-1}, t^{(n)}_{k-1}+r})$. So, by a strong version of Calculus Fundamental Theorem (see [1]), we have

$$F({}_nY_{t^{(n)}_{k-1}, t^{(n)}_k}) - F({}_nY_{t^{(n)}_{k-1}}) = \int_{t^{(n)}_{k-1}}^{t^{(n)}_k} \Delta_t F({}_nY_{t^{(n)}_{k-1}, r})\, dr$$

Let $i_n : [0,T] \to \{1, ..., k_n\}$ be $i_n(s) = k$ if $s \in [t^{(n)}_{k-1}, t^{(n)}_k)$ and $i_n(T) = k_n$. Hence,

$$C_n = \sum_{k=1}^{k_n} F({}_nY_{t^{(n)}_{k-1}, t^{(n)}_k}) - F({}_nY_{t^{(n)}_{k-1}}) = \int_0^T \Delta_t F({}_nY_{t^{(n)}_{i_n(r)-1}, r})\, dr$$

Notice that ${}_nY_{t^{(n)}_{i_n(r)-1}, r}$ is a piecewise constant approximation from below of X_r and X_r is continuous and defined in a compact set $[0,r]$. Hence, Lemma 4.4.1 give us $d_\infty({}_nY_{t^{(n)}_{i_n(r)-1}, r}, X_r) \to 0$. Futhermore, it is immediate that $({}_nY_{t^{(n)}_{i_n(r)-1}, r})_{r\in[0,T]}$ is

adapted.

Since $\Delta_t F$ is continuous, we have $\sup_{r\in[0,T]} \|\Delta_t F({}_nY_{t^{(n)}_{i_n(r)-1},r}) - \Delta_t F(X_r)\| \to 0$ from Λ-Lemma (Lemma 4.2.8). Therefore, Lemma 4.4.2 implies

$$C_n = \int_0^T \Delta_t F({}_nY_{t^{(n)}_{i_n(r)-1},r})\, dr \to \int_0^T \Delta_t F(X_r)\, dr$$

This convergence is pointwise, consequently, is almost surely and moreover in probability.

Consequently,

$$\begin{aligned} F(X_T) - F(X_0) &= \lim_{n\to\infty} A_n + \lim_{n\to\infty} B_n + \lim_{n\to\infty} C_n \\ &= \int_0^T \Delta_t F(X_t)\, dt + \int_0^T \Delta_x F(X_t)\, dx_t + \frac{1}{2}\int_0^T \text{tr}(\Delta_{xx} F(X_t)\, d\langle x\rangle_t)\ \mathbb{P}\text{- a.s.} \end{aligned}$$

as desired. □

Example 4.4.7. *For the functional of Example 4.3.5, it is easy to see that the Itô's Functional Formula reduces to classical Itô's Formula.*

Example 4.4.8. *For the functional of Example 4.3.6, we note, by Itô's Functional Formula, that*

$$F(X_T) - F(X_0) = \int_0^T X_t\, dt$$

what was already obviously to see since $F(X_t) = \int_0^t X_t\, dt$.

5 Functional Ito Calculus: Advanced

In this chapter we extend the classical objects of stochastic analysis—stochastic differential equations, infinitesimal generators, Feynman–Kac formulae, and martingale representation—to the path-dependent setting. The functional Itô formula established in Theorem 4.4.5 is the main tool. We shall see that many classical results admit natural analogues when the coefficients and the functionals depend on the whole past trajectory, rather than only on the current state.

5.1 Functional stochastic differential equations

We consider stochastic differential equations whose coefficients depend on the entire path of the solution up to the current time. This is the natural extension of the classical SDE to the functional framework.

Let $b : \Lambda \to \mathbb{R}^m$ and $\sigma : \Lambda \to \mathbb{R}^{m\times d}$ be continuous functionals. We call them the functional drift and functional diffusion coefficients, respectively. Given an initial condition $\xi \in \mathbb{R}^d$, we are interested in the Functional Stochastic Differential Equation (FSDE)

$$dx_t = b(X_t)dt + \sigma(X_t)dW_t, \qquad x_0 = \xi,$$

where $(W_t)_{t\geq 0}$ is a standard d-dimensional Brownian motion. In integral form,

$$x_t = \xi + \int_0^t b(X_s)\, ds + \int_0^t \sigma(X_s)\, dW_s, \qquad 0 \leq t \leq T. \tag{5.1}$$

Recall that here X_t denotes the path of the continuous semimartingale $(x_t)_{t\geq 0}$ up to time t, i.e. $X_t = (x_s)_{0\leq s\leq t} \in C([0,t],\mathbb{R}^d)$. An adapted continuous process $x = (x_t)_{0\leq t\leq T}$ is called a strong solution of (5.1) if

- $\displaystyle\int_0^T \|b(X_t)\|\, dt < \infty$ and $\displaystyle\int_0^T \|\sigma(X_t)\|^2\, dt < \infty$ almost surely;
- the integral equation (5.1) holds $\mathbb{P}$-a.s. for every $t \in [0,T]$.

Remark 5.1.1. *Since b and σ are continuous and $t \mapsto X_t$ is continuous in the metric d_Λ (Proposition 4.1.5), the processes $(b(X_t))_t$ and $(\sigma(X_t))_t$ are continuous, hence progressively measurable. Therefore the stochastic integrals in* (5.1) *are well-defined.*

We impose the following standard Lipschitz and growth conditions, which are the functional analogues of the classical assumptions.

Assumption 5.1.2. *There exists a constant $K > 0$ such that for all $\omega_t, \nu_t \in C([0,t],\mathbb{R}^d)$ with $t \in [0,T]$,*

$$\|b(\omega_t) - b(\nu_t)\| + \|\sigma(\omega_t) - \sigma(\nu_t)\| \le K\|\omega_t - \nu_t\|_\infty,$$
$$\|b(\omega_t)\|^2 + \|\sigma(\omega_t)\|^2 \le K^2(1 + \|\omega_t\|_\infty^2).$$

We now state the existence and uniqueness theorem.

Theorem 5.1.3 (Existence and uniqueness for FSDEs)**.** *Under Assumption 5.1.2, for every initial condition $\xi \in \mathbb{R}^d$, the functional stochastic differential equation* (5.1) *admits a unique strong solution $x \in L^2(\Omega; C([0,T],\mathbb{R}^d))$.*

Proof. The proof follows the classical Picard iteration scheme, adapted to the path space. We work on the Banach space $\mathcal{H}_T := L^2(\Omega; C([0,T],\mathbb{R}^d))$ equipped with the norm

$$\|x\|_{\mathcal{H}_T} := \left(\mathbb{E}\left[\sup_{0\le t\le T}\|x_t\|^2\right]\right)^{1/2}.$$

Define the operator Φ on $\mathcal{H}_T$ by

$$\Phi(x)_t := \xi + \int_0^t b(X_s)\,ds + \int_0^t \sigma(X_s)\,dW_s, \qquad 0 \le t \le T.$$

We show that Φ is a contraction on $\mathcal{H}_T$ for sufficiently small T; the general case follows by iteration on successive intervals.

Let $x, y \in \mathcal{H}_T$ and denote by X_t, Y_t their respective paths. For each $t \le T$,

$$\Phi(x)_t - \Phi(y)_t = \int_0^t \big(b(X_s) - b(Y_s)\big)\,ds + \int_0^t \big(\sigma(X_s) - \sigma(Y_s)\big)\,dW_s.$$

Using the elementary inequality $(a+b)^2 \le 2a^2 + 2b^2$, the Lipschitz property, Doob's

maximal inequality (Proposition 2.2.1), and the Itô isometry, we obtain

$$\begin{aligned}
\mathbb{E}\left[\sup_{0\le t\le T}\|\Phi(x)_t-\Phi(y)_t\|^2\right] &\le 2\,\mathbb{E}\left[\sup_{t\le T}\left\|\int_0^t (b(X_s)-b(Y_s))\,ds\right\|^2\right] \\
&\quad + 2\,\mathbb{E}\left[\sup_{t\le T}\left\|\int_0^t (\sigma(X_s)-\sigma(Y_s))\,dW_s\right\|^2\right] \\
&\le 2T\,\mathbb{E}\left[\int_0^T \|b(X_s)-b(Y_s)\|^2\,ds\right] \\
&\quad + 8\,\mathbb{E}\left[\int_0^T \|\sigma(X_s)-\sigma(Y_s)\|^2\,ds\right] \\
&\le 2(T+4)K^2\int_0^T \mathbb{E}\left[\sup_{0\le r\le s}\|x_r-y_r\|^2\right] ds \\
&\le 2(T+4)K^2T\sup_{0\le s\le T}\mathbb{E}\left[\sup_{0\le r\le s}\|x_r-y_r\|^2\right].
\end{aligned}$$

Thus,

$$\|\Phi(x)-\Phi(y)\|_{\mathcal{H}_T}^2 \le C_T\|x-y\|_{\mathcal{H}_T}^2,$$

where $C_T = 2K^2T(T+4)$. For $T < (2K^2(T+4))^{-1}$, $C_T < 1$, so Φ is a contraction. By the Banach fixed point theorem, there exists a unique fixed point in $\mathcal{H}_T$, which is the unique strong solution. The growth condition ensures that Φ maps $\mathcal{H}_T$ into itself and that the solution has finite second moment. The extension to arbitrary T is done by concatenation of solutions on small intervals. □

Remark 5.1.4. *The proof also shows that the solution depends continuously on the initial condition ξ in the L^2-sense. Moreover, if the coefficients are only locally Lipschitz, one can use a standard localisation argument to obtain existence up to a possible explosion time.*

5.2 The infinitesimal generator and the martingale property

Before studying the functional Feynman–Kac formula, we introduce the natural analogue of the infinitesimal generator for path-dependent functionals. Let $(x_t)_{t\ge0}$ be a continuous semimartingale satisfying the FSDE

$$dx_t = b(X_t)\,dt + \sigma(X_t)\,dW_t,$$

where $b : \Lambda \to \mathbb{R}^m$ and $\sigma : \Lambda \to \mathbb{R}^{m\times d}$ are continuous functionals. For $F \in C^{1,2}(\Lambda, \mathbb{R})$, define the **functional generator** $\mathcal{L}$ by

$$\mathcal{L}F(\omega_t) := \Delta_t F(\omega_t) + \Delta_x F(\omega_t) b(\omega_t) + \frac{1}{2} \operatorname{tr}\big(\Delta_{xx} F(\omega_t) a(\omega_t)\big), \tag{5.2}$$

where $a(\omega_t) = \sigma(\omega_t)\sigma(\omega_t)^\top \in \mathbb{R}^{m\times m}$. This operator is the path-dependent counterpart of the classical second-order elliptic operator (3.4). Note that $\Delta_t F$ accounts for the "time" variation of the functional along a frozen path, while the remaining terms describe the effect of the diffusion's drift and volatility.

The following lemma is a direct consequence of the functional Itô formula.

Lemma 5.2.1 (Martingale property)**.** *Let $F \in C^{1,2}(\Lambda, \mathbb{R})$ and assume that the processes $\Delta_t F(X_t)$, $\Delta_x F(X_t)$ and $\Delta_{xx} F(X_t)$ are sufficiently integrable so that all terms below are well-defined. Then the process*

$$M_t := F(X_t) - \int_0^t \mathcal{L}F(X_s)\, ds, \qquad 0 \le t \le T,$$

is a continuous local martingale. If, in addition,

$$\mathbb{E}\left[\int_0^T \|\Delta_x F(X_s)\sigma(X_s)\|^2\, ds\right] < \infty,$$

then M is a square-integrable martingale.

Proof. Apply the functional Itô formula (Theorem 4.4.5) to $F(X_t)$:

$$\begin{aligned} F(X_t) = F(X_0) &+ \int_0^t \Delta_t F(X_s)\, ds \\ &+ \int_0^t \Delta_x F(X_s)\, dx_s + \frac{1}{2}\int_0^t \operatorname{tr}\big(\Delta_{xx} F(X_s)\, d\langle x\rangle_s\big). \end{aligned}$$

Since $dx_s = b(X_s)\, ds + \sigma(X_s)\, dW_s$ and $d\langle x\rangle_s = a(X_s)\, ds$, substituting and collecting the Lebesgue-integral terms gives

$$F(X_t) = F(X_0) + \int_0^t \mathcal{L}F(X_s)\, ds + \int_0^t \Delta_x F(X_s)\sigma(X_s)\, dW_s.$$

Therefore

$$M_t = F(X_0) + \int_0^t \Delta_x F(X_s)\sigma(X_s)\, dW_s,$$

which is a stochastic integral with respect to Brownian motion. Hence it is a local martingale; under the stated square-integrability condition it is a martingale. □

Remark 5.2.2. *The generator $\mathcal{L}$ is sometimes called the **functional Dynkin operator**.*

When F depends only on the current value, i.e. $F(\omega_t) = f(t, \omega_t(t))$, formula (5.2) *reduces to the classical generator $\mathcal{L}f = \partial_t f + b \cdot \nabla_x f + \frac{1}{2}\operatorname{tr}(a \nabla^2_{xx} f)$.*

5.3 Functional Feynman–Kac formula

We now consider a path-dependent partial differential equation of the form

$$\begin{cases} \mathcal{L}u(\omega_t) - c(\omega_t)u(\omega_t) = -f(\omega_t), & \omega_t \in C([0,t],\mathbb{R}^d),\ 0 \le t < T, \\ u(\omega_T) = g(\omega_T), & \omega_T \in C([0,T],\mathbb{R}^d), \end{cases} \tag{5.3}$$

where $c, f : \Lambda \to \mathbb{R}$ are continuous functionals and $g : \Lambda \to \mathbb{R}$ is a given terminal functional. Here the "time derivative" is the horizontal derivative $\Delta_t u$ and $\mathcal{L}$ acts on the path variable at time t. Equation (5.3) is the functional analogue of the classical Feynman–Kac PDE.

To give a probabilistic representation, we need the solution of the FSDE starting from an arbitrary initial path $\omega_t \in C([0,t],\mathbb{R}^d)$ at time t. For $s \in [t,T]$, denote by $X_s^{t,\omega}$ the unique strong solution of

$$X_s^{t,\omega} = \omega_t(s) \quad \text{for } s \le t, \qquad dX_s^{t,\omega} = b(X_s^{t,\omega})\,ds + \sigma(X_s^{t,\omega})\,dW_s \quad \text{for } s > t.$$

For $s > t$, $X_s^{t,\omega}$ denotes the whole trajectory restricted to $[0,s]$. The well-posedness of this "functional initial value problem" follows from Theorem 5.1.3 applied on the interval $[t,T]$ with the initial path ω_t, since the coefficients satisfy the Lipschitz condition on the space of continuous paths.

Theorem 5.3.1 (Functional Feynman–Kac formula)**.** *Assume that b, σ satisfy Assumption 5.1.2 and that c, f, g are continuous functionals with at most linear growth in $\|\omega_t\|_\infty$. Suppose $u \in C^{1,2}(\Lambda,\mathbb{R})$ is a solution of the functional PDE* (5.3)*. Then, for every $\omega_t \in C([0,t],\mathbb{R}^d)$,*

$$u(\omega_t) = \mathbb{E}\left[g(X_T^{t,\omega}) e^{-\int_t^T c(X_s^{t,\omega})\,ds} + \int_t^T f(X_s^{t,\omega}) e^{-\int_t^s c(X_r^{t,\omega})\,dr}\,ds \right], \tag{5.4}$$

where the expectation is taken under the probability measure of the Brownian motion W.

Proof. For brevity we drop the superscript (t,ω) and write X_s for $X_s^{t,\omega}$. Define the process

$$Y_s := u(X_s) e^{-\int_t^s c(X_r)\,dr}, \qquad s \in [t,T].$$

We apply the functional Itô formula to Y_s. First note that the exponential factor is a finite-

variation process with differential

$$d\left(e^{-\int_t^s c(X_r)\,dr}\right) = -c(X_s)e^{-\int_t^s c(X_r)\,dr}\,ds,$$

because $r \mapsto c(X_r)$ is continuous (by continuity of c and of $r \mapsto X_r$). Using the product rule for functional derivatives (Proposition 4.3.3) and the functional Itô formula,

$$\begin{aligned} dY_s &= e^{-\int_t^s c\,dr}\,du(X_s) + u(X_s)\,d\left(e^{-\int_t^s c\,dr}\right) \\ &= e^{-\int_t^s c\,dr}\left(\mathcal{L}u(X_s)\,ds + \Delta_x u(X_s)\sigma(X_s)\,dW_s\right) \\ &\quad - c(X_s)u(X_s)e^{-\int_t^s c\,dr}\,ds \\ &= e^{-\int_t^s c\,dr}\left(\mathcal{L}u(X_s) - c(X_s)u(X_s)\right)ds \\ &\quad + e^{-\int_t^s c\,dr}\Delta_x u(X_s)\sigma(X_s)\,dW_s. \end{aligned}$$

Using the PDE $\mathcal{L}u - cu = -f$, we get

$$dY_s = -f(X_s)e^{-\int_t^s c\,dr}\,ds + e^{-\int_t^s c\,dr}\Delta_x u(X_s)\sigma(X_s)\,dW_s.$$

Integrating from t to T,

$$Y_T - Y_t = -\int_t^T f(X_s)e^{-\int_t^s c\,dr}\,ds + \int_t^T e^{-\int_t^s c\,dr}\Delta_x u(X_s)\sigma(X_s)\,dW_s.$$

Taking expectations and using that the stochastic integral is a martingale (which follows from the growth assumptions and the boundedness of the exponential factor), we obtain

$$\mathbb{E}[Y_T] = Y_t - \mathbb{E}\left[\int_t^T f(X_s)e^{-\int_t^s c\,dr}\,ds\right].$$

Since $Y_t = u(\omega_t)$ (the exponential is 1 at $s = t$) and $Y_T = u(X_T)e^{-\int_t^T c\,dr} = g(X_T)e^{-\int_t^T c\,dr}$, the representation (5.4) follows. □

Remark 5.3.2. *In the homogeneous case $f \equiv 0$ and $c \equiv 0$, formula* (5.4) *reduces to*

$$u(\omega_t) = \mathbb{E}\big[g(X_T^{t,\omega})\big],$$

which is the expected terminal payoff. The corresponding PDE is $\mathcal{L}u = 0$ with terminal condition $u = g$ on Λ.

Remark 5.3.3. *The proof also shows uniqueness of the solution to* (5.3) *within the class of functions satisfying the growth assumptions, because the representation* (5.4) *is explicit in terms of the data f, g, c.*

5.4 Martingale representation theorem for functional diffusions

We now turn to the representation of martingales adapted to the filtration generated by the solution x of the FSDE. Assume that the diffusion matrix σ is invertible (or at least that the covariance matrix $a = \sigma\sigma^\top$ is uniformly positive definite), so that the Brownian motion W can be recovered from x. More precisely, let $(\mathcal{F}_t)_{t\geq 0}$ be the natural filtration of x (which equals that of W if σ is invertible). We want to show that every square-integrable $\mathcal{F}_T$-martingale M can be represented as

$$M_t = M_0 + \int_0^t \phi_s \, dW_s$$

for some predictable process ϕ (with values in $\mathbb{R}^d$) satisfying $\mathbb{E}\int_0^T \|\phi_s\|^2 \, ds < \infty$. In the path-dependent setting, the integrand ϕ will itself be a functional of the path, i.e. $\phi_s = \Phi(X_s)$ for some functional $\Phi : \Lambda \to \mathbb{R}^d$.

The theorem below is the functional analogue of the classical martingale representation theorem (Theorem 3.4.4).

Theorem 5.4.1 (Functional martingale representation)**.** *Let $(x_t)_{t\geq 0}$ be the strong solution of the FSDE* (5.1) *with coefficients $b, \sigma \in C^{1,2}(\Lambda)$ (in the sense of functional derivatives) and suppose σ is invertible with bounded inverse. Then, for every square-integrable $\mathcal{F}_T$-measurable random variable F (with $\mathbb{E}|F|^2 < \infty$), there exists a unique predictable process $(\phi_t)_{0\leq t\leq T}$ such that*

$$F = \mathbb{E}[F] + \int_0^T \phi_t \, dW_t. \tag{5.5}$$

Equivalently, the martingale $M_t := \mathbb{E}[F \mid \mathcal{F}_t]$ admits the representation

$$M_t = \mathbb{E}[F] + \int_0^t \phi_s \, dW_s, \qquad 0 \leq t \leq T.$$

Proof. We give a proof by approximating F with smooth cylindrical functionals of the path. The idea is standard: first consider functionals of the form

$$F = f(x_{t_1}, x_{t_2}, \ldots, x_{t_n})$$

with $f \in C_b^\infty(\mathbb{R}^{nd})$ and $0 \leq t_1 < \cdots < t_n \leq T$. Such functionals are dense in $L^2(\mathcal{F}_T)$ (by the monotone class theorem, since the increments of Brownian motion generate the

filtration). For such a cylindrical functional, define a path-dependent functional

$$G(\omega_T) := f(\omega_T(t_1), \dots, \omega_T(t_n)).$$

Then G is continuous on Λ (by continuity of evaluation at fixed times) and is $C^{1,2}$ in the functional sense. Indeed, its horizontal derivative is zero because G depends on ω_T only through its values at fixed times, and its vertical derivative is given by

$$\Delta_x G(\omega_T) = \sum_{i=1}^{n} \nabla_i f(\omega_T(t_1), \dots, \omega_T(t_n))\, \mathbf{1}_{\{t_i = T\}},$$

where ∇_i denotes the gradient with respect to the i-th block of variables. In particular, $\Delta_x G(\omega_T)$ is non-zero only if one of the times t_i equals T; otherwise it is zero.

Applying the functional Itô formula (Theorem 4.4.5) to $G(X_t)$ for $t \le T$, we obtain

$$G(X_T) = G(X_0) + \int_0^T \mathcal{L}G(X_s)\, ds + \int_0^T \Delta_x G(X_s)\sigma(X_s)\, dW_s.$$

Taking conditional expectation given $\mathcal{F}_t$ and using the martingale property from Lemma 5.2.1, we get

$$\mathbb{E}[G(X_T)|\mathcal{F}_t] = G(X_t).$$

Thus,

$$G(X_t) = G(X_0) + \int_0^t \mathcal{L}G(X_s)\, ds + \int_0^t \Delta_x G(X_s)\sigma(X_s)\, dW_s.$$

But from the functional Itô formula applied directly up to t, this is consistent. Since $G(X_T) = F$, we have

$$\mathbb{E}[F|\mathcal{F}_t] = \mathbb{E}[F] + \int_0^t \Delta_x G(X_s)\sigma(X_s)\, dW_s.$$

Hence $\phi_s^{(G)} = \Delta_x G(X_s)\sigma(X_s)$ provides the representation for this particular F.

For a general $F \in L^2(\mathcal{F}_T)$, choose a sequence F_n of cylindrical smooth functionals converging to F in L^2. For each n, by the above argument, there exists $\phi^{(n)}$ such that

$$\mathbb{E}[F_n|\mathcal{F}_t] = \mathbb{E}[F_n] + \int_0^t \phi_s^{(n)}\, dW_s.$$

The martingale parts converge in L^2 because the conditional expectations converge in L^2. By the Itô isometry, the sequence $\phi^{(n)}$ is Cauchy in $L^2(\Omega \times [0,T]; \mathbb{R}^d)$. Therefore there exists a limiting predictable process ϕ satisfying

$$\mathbb{E}[F|\mathcal{F}_t] = \mathbb{E}[F] + \int_0^t \phi_s\, dW_s.$$

Setting $t = T$ yields (5.5). Uniqueness follows from the Itô isometry: if ϕ and $\tilde{\phi}$ both represent the same martingale, then $\int_0^T (\phi_s - \tilde{\phi}_s) dW_s = 0$ a.s., so by the isometry $\mathbb{E} \int_0^T \|\phi_s - \tilde{\phi}_s\|^2 ds = 0$, hence $\phi = \tilde{\phi}$ in L^2. □

Remark 5.4.2. *In the proof above we used the fact that the filtration generated by x coincides with that generated by W. This holds because σ is invertible: one can solve*

$$dW_t = \sigma(X_t)^{-1}(dx_t - b(X_t)\, dt),$$

which expresses W as a functional of x. Conversely, x is a functional of W by the FSDE. Thus $\mathcal{F}_t^x = \mathcal{F}_t^W$ (after completing the filtrations).

Remark 5.4.3. *The functional martingale representation theorem is a powerful tool for hedging path-dependent derivatives in mathematical finance. The integrand ϕ_t gives the amount of the risky asset to hold at time t in a replicating portfolio, and it depends on the whole past trajectory of the underlying asset through the functional $\Delta_x G$.*

6 Itô Functional Calculus: Applications

In this chapter we illustrate the practical utility of the functional Itô calculus developed in the previous chapters. The classical Markovian setting, while powerful, is insufficient for many modern problems in finance, engineering, and physics, where the dependence on the past trajectory is intrinsic. The functional framework provides a rigorous mathematical foundation for analyzing these problems. We present three prototypical applications: pricing and hedging of path-dependent derivatives, stochastic control with memory, and functional stochastic differential equations with delay.

6.1 Pricing and hedging of path-dependent derivatives

One of the most direct and celebrated applications of the functional Itô calculus is in mathematical finance, particularly for exotic options whose payoff depends on the entire history of the underlying asset price, rather than just its terminal value.

6.1.1 Functional pricing PDE

Let $(S_t)_{t\geq 0}$ be the price of a risky asset, modeled under a risk-neutral measure $\mathbb{P}$ as

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t,$$

where r is the constant risk-free rate, $\sigma > 0$ the volatility, and (W_t) a standard Brownian motion. In our functional notation, we set $x_t = S_t$ and X_t as the path up to time t.

Consider a path-dependent European claim with payoff at maturity T given by a continuous functional $g : \Lambda \to \mathbb{R}$. Examples include:

- **Asian option**: $g(X_T) = \left(\dfrac{1}{T}\displaystyle\int_0^T S_s\,ds - K\right)_+$.
- **Lookback option**: $g(X_T) = \left(\displaystyle\max_{0\leq s\leq T} S_s - K\right)_+$.
- **Barrier option**: $g(X_T) = (S_T - K)_+ \mathbf{1}_{\{\min_{0\leq s\leq T} S_s > B\}}$.

Under standard no-arbitrage theory, the price at time t given the observed past path $\omega_t \in C([0,t],\mathbb{R}_+)$ is

$$V(t,\omega_t) := \mathbb{E}\left[e^{-r(T-t)}g(X_T^{t,\omega})\right],$$

where $X_T^{t,\omega}$ denotes the unique strong solution of the FSDE starting from the path ω_t at time t. Applying the functional Feynman–Kac formula (Theorem 5.3.1) with $c \equiv r$ (constant discounting) and $f \equiv 0$, we obtain the functional pricing PDE:

$$\begin{cases} \mathcal{L}V(\omega_t) - rV(\omega_t) = 0, & \omega_t \in C([0,t],\mathbb{R}_+),\ 0 \le t < T, \\ V(\omega_T) = g(\omega_T), & \omega_T \in C([0,T],\mathbb{R}_+), \end{cases} \tag{6.1}$$

where the functional generator $\mathcal{L}$ is given by (5.2). Using the specific dynamics $b(\omega_t) = r\omega_t(t)$ and $\sigma(\omega_t) = \sigma\omega_t(t)$, we have

$$\mathcal{L}V(\omega_t) = \Delta_t V(\omega_t) + r\omega_t(t)\Delta_x V(\omega_t) + \frac{1}{2}\sigma^2\omega_t(t)^2\Delta_{xx}V(\omega_t).$$

Equation (6.1) is the path-dependent analogue of the classical Black–Scholes PDE. While classical PDEs are solved on a finite-dimensional domain (t,S), the functional PDE (6.1) is an infinite-dimensional equation defined on the path space Λ.

Remark 6.1.1. *In practice, one often reduces the infinite-dimensional problem to a finite-dimensional one by exploiting the specific structure of the payoff. For example, for an Asian option, the price depends on the pair (S_t, I_t), where $I_t = \int_0^t S_s\,ds$. In this case, the functional reduces to a classical function of (t, S_t, I_t), and (6.1) collapses to the well-known two-dimensional PDE for Asian options.*

6.1.2 Hedging and the functional delta

A fundamental problem in finance is the replication of a contingent claim. The martingale representation theorem for functional diffusions (Theorem 5.4.1) provides the exact hedging strategy. Indeed, the discounted price process $(e^{-rt}V(X_t))_{t\in[0,T]}$ is a martingale under the risk-neutral measure. Applying the functional Itô formula and using the pricing PDE (6.1), we get

$$d\left(e^{-rt}V(X_t)\right) = e^{-rt}\Delta_x V(X_t)\sigma X_t\,dW_t.$$

Hence, by Theorem 5.4.1, the replicating portfolio consists of:

- $\Delta_x V(X_t)$ shares of the risky asset (the functional delta);
- the remaining amount invested in the risk-free asset to self-finance the portfolio.

Unlike the classical delta $\partial_S V(t, S_t)$, the functional delta $\Delta_x V(X_t)$ depends on the entire past trajectory of the asset price. This is a crucial feature for hedging exotic options, where the sensitivity to the past (e.g., the running average or the running maximum) must be taken into account.

6.2 Stochastic optimal control with path dependence

Many optimal control problems in economics, operations research, and engineering involve costs or constraints that depend on the history of the state process. The functional Itô calculus provides the natural framework for such problems.

6.2.1 The control problem

Consider a controlled functional stochastic differential equation

$$dx_t = b(X_t, u_t)\, dt + \sigma(X_t, u_t)\, dW_t, \qquad x_0 = \xi,$$

where the control process $(u_t)_{t\geq 0}$ takes values in a compact set $U \subset \mathbb{R}^m$ and is adapted to the filtration. The cost functional over the horizon $[t, T]$ is

$$J(t, \omega_t; u) := \mathbb{E}\left[\int_t^T f(X_s, u_s)\, ds + g(X_T) \;\middle|\; X_t = \omega_t \right],$$

where $f : \Lambda \times U \to \mathbb{R}$ is the running cost and $g : \Lambda \to \mathbb{R}$ is the terminal cost. The value function is defined as

$$V(t, \omega_t) := \inf_u J(t, \omega_t; u).$$

6.2.2 Functional Hamilton-Jacobi-Bellman (HJB) equation

By applying the dynamic programming principle in the path-dependent setting and using the functional Itô formula, we derive the functional HJB equation:

$$\begin{cases} \inf\limits_{u\in U} \{\mathcal{L}^u V(\omega_t) + f(\omega_t, u)\} = 0, & \omega_t \in C([0,t], \mathbb{R}^d),\ 0 \leq t < T, \\ V(T, \omega_T) = g(\omega_T), & \omega_T \in C([0,T], \mathbb{R}^d), \end{cases} \tag{6.2}$$

where $\mathcal{L}^u$ denotes the functional generator associated with the controlled dynamics, i.e.,

$$\mathcal{L}^u V(\omega_t) := \Delta_t V(\omega_t) + \Delta_x V(\omega_t) b(\omega_t, u) + \frac{1}{2} \operatorname{tr}\left(\Delta_{xx} V(\omega_t) \sigma(\omega_t, u) \sigma(\omega_t, u)^\top \right).$$

The verification theorem for this functional HJB equation states that if a sufficiently regular functional $V \in C^{1,2}(\Lambda)$ satisfies (6.2) and suitable growth conditions, then V

coincides with the value function, and the minimizer $u^*(\omega_t)$ of the Hamiltonian provides the optimal feedback control. This generalizes the classical Markovian HJB theory to systems with memory.

Remark 6.2.1. *When the dynamics and the cost are Markovian (i.e., depend only on the current state x_t), equation (6.2) reduces to the standard HJB equation. However, in many practical applications, such as optimal execution in finance with transient price impact or inventory control with backlog costs, the dependence on past orders or past prices is essential, making the functional HJB formulation indispensable.*

6.3 Stochastic functional differential equations (SFDEs)

Another important class of applications involves stochastic systems with delay or memory, often modeled by stochastic functional differential equations. A typical SFDE has the form

$$dx_t = b(X_t)\,dt + \sigma(X_t)\,dW_t,$$

which is exactly the FSDE studied in Section 5.1. In many physical and biological systems, the coefficients b and σ depend on the entire past trajectory through, for example, delays:

$$b(X_t) = \tilde{b}(x_t, x_{t-\tau}), \qquad \sigma(X_t) = \tilde{\sigma}(x_t, x_{t-\tau}),$$

for some fixed lag $\tau > 0$. These are special cases of our general functional framework where b and σ are continuous on the path space.

Example 6.3.1 (Stochastic delay differential equation)**.** *Consider the equation*

$$dx_t = -a x_{t-\tau}\,dt + \sigma x_t\,dW_t, \qquad t \geq 0,$$

with a given initial path on $[-\tau, 0]$. This system arises in population dynamics with delayed regulation and in finance with delayed feedback. Under appropriate Lipschitz conditions on the path space, Theorem 5.1.3 guarantees the existence and uniqueness of a strong solution. Moreover, the infinitesimal generator $\mathcal{L}$ can be explicitly computed in terms of the delayed state, allowing one to study stability and ergodicity using the functional Feynman–Kac formula (Theorem 5.3.1).

The functional framework also provides a rigorous setting for numerical methods. For instance, the functional Itô formula suggests pathwise Taylor expansions, which can be used to construct high-order numerical schemes for SFDEs, generalizing the classical stochastic Runge–Kutta methods.

6.4 Summary and outlook

In this chapter we have demonstrated the versatility of the functional Itô calculus through three important applications:

1. **Finance**: The pricing and hedging of path-dependent options are naturally formulated via functional PDEs and martingale representation, with the functional delta providing the exact replication strategy.

2. **Control**: Stochastic optimal control with memory leads to a functional HJB equation, which generalizes the classical dynamic programming theory.

3. **Systems with memory**: Stochastic functional differential equations, including delay equations, are rigorously analyzed using the existence and uniqueness theorems for FSDEs.

Beyond these, the theory finds applications in:

- **Path-dependent risk measures**: Coherent risk measures that depend on the entire trajectory of a portfolio (e.g., drawdown constraints) can be studied using functional derivatives.

- **Statistical inference for path-dependent models**: The functional Feynman–Kac formula enables the computation of likelihood ratios for models with memory.

- **Mean-field games with common noise**: When the interaction depends on the empirical distribution of past trajectories, functional calculus provides the necessary analytical tools.

The development of efficient numerical algorithms—such as functional quantization, deep learning approximations of the functional PDEs, and Monte Carlo methods for FSDEs—remains a vibrant area of ongoing research.

7 Conclusion

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